

# Limited-Memory Renewal Contracts: Harmonic Coarsening and Certified Joint Design

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**Abstract.** A renewal provider must select a limited command alphabet and allocate an accepted obligation across heterogeneous histories. We develop contract-preserving joint optimization for a finite catalog with opening charges, participation caps, and realization ceilings. An eligibility-aware two-sided price-path oracle supplies globally valid bounds on target boxes. Guarded aggregation reduces the history dimension, and weighted harmonic service curvatures tighten the reduced model without adding target coordinates. We prove dominance over minimum-curvature aggregation and construct an optimal within-group lift satisfying every original constraint. A square-root curvature partition gives a second-order cost-dispersion bound and improves the accuracy-dependent dimension of the variable-history additive scheme. For bounded eligibility complexity and primitive scales, fixed-accuracy complexity is polynomial in the history count and catalog size; accuracy dependence remains potentially large. Independent checkers verify reduced upper bounds and original policy feasibility. A paired 24-case cost-heterogeneity study includes up to 256 histories, exact joint references, numerical mixed-integer comparisons, and retained unresolved cases. Canonical contractual allocation, continuous quadratic design, Monge acceleration, and budgeted compression remain under their original assumptions. The method strengthens joint-design certificates without expanding the installed command budget.

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# 1 Introduction

A renewal service provider may observe many histories while its terminal gateway accepts only a small menu of commands. The provider must choose that menu and allocate an accepted service obligation across histories. Participation limits expected service on each history, a separate ceiling limits every realized delivery, and installing a command can incur a fixed charge. These constraints make the menu and the promised service targets interdependent. Optimizing commands for prescribed targets does not solve the joint design problem.

We develop a joint-design method that preserves the original command budget, charges, aggregate obligation, and individual constraints. Its first component is an exact price oracle on a box of target intervals. A mandatory lower target can make commands after the price-optimal peak indispensable. An increasing prefix accounts for upper-target crossings, a decreasing suffix accounts for lower-target crossings, and their common peak joins two path recurrences. The oracle gives valid upper bounds over every menu in the box, not just over a menu supplied by a heuristic. Adaptive partitioning then yields an interval containing the global joint optimum, with finite termination at every positive additive tolerance.

The second component reduces the history dimension without requiring identical histories. Groups retain the same eligible commands but may differ in caps and costs. Weighted caps preserve aggregate capacity, while a minimum-cap guard preserves feasible commands. We strengthen minimum-curvature aggregation by using weighted harmonic service curvatures. This gives a provably tighter joint upper model without extra target coordinates. Recovery then optimizes each group’s original allocation at its reduced mean, preserving the installed menu, exact obligation, individual caps, and every realization ceiling.

A square-root curvature partition gives a second-order cost-dispersion bound, including zero costs. Together with cap-dispersion control, it improves the accuracy-dependent dimension of the variable-history additive scheme. For bounded eligibility complexity and primitive scales, fixed-accuracy complexity is polynomial in the number of histories and catalog size. The accuracy dependence can still be large. Practical coarse partitions are judged by a checked original-space interval, never by the reduced solver’s completion flag alone. Refinement and the best available original-space witnesses provide a constructive route to tighter certificates.

An independent checker verifies the group construction, recovery identities, original lotteries, price-path inequalities, and the complete target cover. The computational evidence includes distinct histories and a paired cost-heterogeneity ablation of harmonic and minimum-curvature aggregation. It compares certified coarsening with unaggregated adaptive search, the inherited priced-prefix method, exhaustive menu enumeration, and an uneliminated mixed-integer formulation. All declared cases and unresolved intervals are retained, including regimes where cap dispersion dominates the improved cost bound. Separate stress instances show that an exact reduced solution can still leave a substantial original-space gap, and that refinement removes this gap. These synthetic experiments evaluate algorithms and certificates, not an estimated operational population.

**Relation to established methods.** Classical interval quantization and matrix searching underpin the saturated and conditional path algorithms (Wu, 1991; Nielsen and Nock, 2014; Grønlund et al., 2017). Pricing,

constrained paths, and branch-and-bound are established tools (Handler and Zang, 1980; Lawler and Wood, 1966; Lemaréchal, 2001). Spatial branch-and-bound also exploits locally tight relaxations of separable piecewise-linear problems (Hübner et al., 2025). We do not claim those general mechanisms as new. Our two-sided oracle handles a shared, charged, cardinality-constrained menu under heterogeneous eligibility, with an anchor condition that ensures local tightness.

Aggregation error analysis is likewise established. Li and Bertsekas (2025) bound value-function error in aggregated discounted dynamic programming, and Fattahi (2025) studies approximation and error components in jointly designing charging programs and managing a heterogeneous participant population. Our result concerns a different coupling: one installed menu and an exact root obligation must survive aggregation and recovery while every original cap and realization ceiling remains enforceable. The operative ingredients are eligibility-preserving groups, the minimum-cap guard, and a group-mass-preserving lift. The inequalities between arithmetic and harmonic means are classical; see, for example, Liao and Wu (2015). Their use here must additionally preserve the shared menu and every original contract constraint. The associated complexity guarantee does not assume a small number of exactly repeated types.

The broader contractual and continuous-design contributions remain part of the paper. Canonical allocation connects expected and pathwise participation; continuous quadratic design, Monge acceleration, conditional compression, full target nets, and charge-aware recovery retain their original assumptions and proofs. Table 1 separates these antecedents from two-sided joint decomposition and certified response coarsening. The article presents the joint-design argument first; the companion retains the full preceding mathematical chain. The repository computational record carries the intact earlier study narratives and the complete new per-instance measurements.

## 2 Model and Canonical Allocation

A public branch  $j$  is observed with probability  $\pi_j > 0$ , with  $\sum_j \pi_j = 1$ . It has a continuation cap  $b_j \in (0, 1)$  and a convex nondecreasing intermediate-service cost  $h_j$ , with  $h_j(0) = 0$ . The common terminal reward  $f$  is strictly increasing and concave on  $[0, 1]$ . Caps may repeat; write  $\bar{b} = \sum_j \pi_j b_j$ . An encoder selects a terminal command from one installed alphabet, and only its symbol reaches the common terminal gateway. The gateway cannot recover the observed branch, an inherited intermediate tier, or an uncharged shared random seed. Thus  $m$  distinct commands require  $\lceil \log_2 m \rceil$  writable bits; read-only program constants and public-state storage are separate resources. The full institutional mechanism and saturated benchmarks are retained in the companion.

The provider selects a pre-draw tier  $y_j$  and a lottery  $C_j$  on the installed book, with operating payoff  $\mathbb{E}f(C_j) - h_j(y_j)$ . The accepted root obligation is  $B \in [0, \bar{b}]$ . Expected participation restricts the mean obligation, while a realization ceiling restricts every positive-probability draw. With a common risk tolerance  $\delta \geq 0$ ,

$$t_j = y_j + \mathbb{E}C_j \leq b_j, \quad y_j + C_j \leq b_j + \delta \text{ almost surely}, \quad \sum_j \pi_j t_j = B. \quad (1)$$

A book  $c$  has at most  $m$  distinct levels in  $[0, 1]$  and incurs  $\sum_{z \in c} \rho(z)$ , once per selected level. Let  $\mathcal{V}_m^\delta(B; \rho)$

denote maximum operating payoff less this charge. A finite catalog restriction is denoted  $\mathcal{V}_{m,\mathcal{A}}^\delta(B;\rho)$ . The expectation cap is not replaced by the weaker realization ceiling. Taking  $\delta \geq 1$  makes the realization restriction redundant at the canonical optimum below. Taking  $\delta = 0$  gives the full pathwise value, even though that institution initially permits draw-specific intermediate service. Thus these are endpoints of one design problem, not three unrelated extensions.

## 2.1 Eligibility prefixes replace saturated targets

For a strictly ordered book  $c = (c_1, \dots, c_s)$  and branch  $j$ , define

$$c^{j,\delta} = \{z \in c : z \leq b_j + \delta\}, \quad d_j = \max c^{j,\delta},$$

and let  $F_{j,c}^\delta$  be the piecewise-affine interpolant of  $f$  on this prefix, including the singleton interpretation. The prefix depends on the cap and risk ceiling, not on the endogenous target.

**Theorem 2.1** (Canonical allocation under a realization ceiling). *Suppose  $f$  is strictly increasing and concave and  $h_j$  is convex, nondecreasing, and zero at zero. A fixed book is feasible exactly when  $c_1 \leq \min_j b_j$  and  $c_1 \leq B \leq \bar{b}$ . Its optimal branch response at target  $c_1 \leq t \leq b_j$  is*

$$v_{j,c}^\delta(t) = F_{j,c}^\delta(\min\{t, d_j\}) - h_j((t - d_j)_+). \quad (2)$$

It is attained by  $y = (t - d_j)_+$  and an adjacent lottery of mean  $\min(t, d_j)$  in the eligible prefix. Consequently

$$\mathcal{V}_m^\delta(B;\rho) = \max_{\substack{|c| \leq m, c_1 \leq t_j \leq b_j \\ \sum_j \pi_j t_j = B}} \left\{ \sum_j \pi_j v_{j,c}^\delta(t_j) - \sum_{z \in c} \rho(z) \right\}. \quad (3)$$

The fixed-book inner problem is concave. With zero charges,  $\mathcal{V}_m^0(B;0) = V_m^P(B)$  and  $\mathcal{V}_m^\delta(B;0) = V_m^E(B)$  for  $\delta \geq 1$ , at every feasible  $B$ .

*Proof.* Every used terminal level satisfies  $C \leq b_j + \delta$  because  $y \geq 0$ . Writing  $\mu = \mathbb{E}C$ , the payoff is at most  $F_{j,c}^\delta(\mu) - h_j(t - \mu)$ . Both terms are nondecreasing as  $\mu$  increases on  $[c_1, \min(t, d_j)]$ , so this upper bound is maximized at its right endpoint. If  $t \leq d_j$ , the adjacent lottery has  $y = 0$  and all its support lies below  $b_j + \delta$ . If  $t > d_j$ , it is the singleton  $d_j$  and  $y + d_j = t \leq b_j$ . Thus the upper bound is attained without relaxing either constraint. The response is concave: its nonnegative interpolation slopes decrease, and its tail has the nonpositive slopes of  $-h_j(t - d_j)$ . The feasibility conditions follow by using the lowest level and allocating intermediate service; weighted targets fill the entire interval  $[c_1, \bar{b}]$ .

For pathwise participation with possibly draw-specific  $Y$ , all used  $C$  satisfy  $C \leq b_j$ . At fixed  $t = \mathbb{E}(Y+C)$ , concavity of  $f$  on the discrete support and Jensen's inequality for  $h_j$  give the same upper bound with  $\delta = 0$  and  $\mathbb{E}Y = t - \mu$ . The canonical policy attains it and is pathwise feasible. Fresh decoder randomization can first be replaced by its conditional mean for each symbol: this increases terminal reward and preserves all the ceilings under the stipulated absence of an uncharged shared seed. Under expected participation the

same mean argument has no eligibility truncation;  $\delta \geq 1$  includes every level in  $[0, 1]$ . These observations establish the two endpoint identities and both directions of (3).  $\square$

For quadratic primitives (54), chord  $[u, v]$  has slope  $r - q(u + v)/2$ . On branch  $j$  its target capacity is

$$\left[ \min\{b_j, v\} - u \right]_+ \mathbf{1}\{v \leq b_j + \delta\}.$$

Starting at  $t_j = c_1$ , fill these positive slopes in their common decreasing order, followed by zero-cost intermediate capacity and the quadratic tails. A water-level sweep over events  $\gamma_j(b_j - d_j)_+$  fills the remaining targets. It uses  $O(k s + k \log(k + 1))$  rational arithmetic operations, including the certificate

$$\sum_j \pi_j v_{j,c}^\delta(t_j) = \lambda B + \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}^\delta(t) - \lambda t\}. \quad (4)$$

The right-hand branch maxima are independently checked at eligible knots, the cap, and the clipped tail stationary point. This multiplier certifies the allocation for that book; it is not by itself a global codebook certificate. The outer problem is solved next.

All canonical arguments continue to hold for branch-specific ceilings  $\tau_j \geq b_j$  in place of  $b_j + \delta$ : the eligibility prefix is then  $c \cap [0, \tau_j]$ . We write  $\mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho)$  for this generalized catalog value. The new budgeted algorithms and numerical formulation use these heterogeneous ceilings. The retained continuous and institutional results use their stated common- $\delta$  specialization.

### 3 Exact Budgeted Compression at Fixed Targets

We first solve alphabet selection without changing the promised branch targets. This is both an independently meaningful design problem and an oracle for joint design. Let  $\mathcal{A} = (a_1 < \dots < a_N)$ , with charges  $\rho_i \geq 0$ , and fix  $t_j \leq b_j$ ,  $\sum_j \pi_j t_j = B$ , and  $a_1 \leq \min_j t_j$ . Write  $\tau_j \geq b_j$  for the realization ceiling; the renewal specialization is  $\tau_j = b_j + \delta$ . In this section  $v_{j,c}$  denotes (2) with this branch-specific ceiling, and  $R(c) = \sum_{a_i \in c} \rho_i$ .

For two candidate levels  $u < v$ , define the target-crossing edge and stopping reward

$$A_t(u, v) = \sum_{j: u \leq t_j < v} \pi_j \begin{cases} f(u) + \frac{t_j - u}{v - u} \{f(v) - f(u)\}, & v \leq \tau_j, \\ f(u) - h_j(t_j - u), & v > \tau_j, \end{cases} \quad (5)$$

$$Z_t(u) = \sum_{j: t_j \geq u} \pi_j \{f(u) - h_j(t_j - u)\}. \quad (6)$$

The eligibility test is on  $v$ , not on its mean with  $u$ . If it fails, a branch uses the lower level and a fixed tier  $t_j - u$ ; averaging an ineligible realization into a feasible mean would not implement the contract. Unlike the priced recurrence, (5) has no price-admissibility restriction on an arc.

**Theorem 3.1** (Conditional memory–charge frontier). *For any feasible target vector  $t$ , define*

$$F_\ell(t) = \max_{\substack{c \subseteq \mathcal{A}, |c|=\ell \\ c_1 \leq \min_j t_j}} \left\{ \sum_j \pi_j v_{j,c}(t_j) - R(c) \right\}, \quad P_s(t) = \max_{1 \leq \ell \leq s} F_\ell(t).$$

*Infeasible exact cardinalities have value  $-\infty$ . Set  $H_1(i) = Z_t(a_i)$  and*

$$H_\ell(i) = \max_{v > i} \{A_t(a_i, a_v) - \rho_v + H_{\ell-1}(v)\}. \quad (7)$$

*Then  $F_\ell(t) = \max_{a_i \leq \min_j t_j} \{H_\ell(i) - \rho_i\}$ . All frontiers through budget  $M \leq N$  and their feasible policies are computable in  $O((k + M)N^2)$  arithmetic operations and  $O(N^2 + MN)$  storage, given constant-time evaluations of  $f$  and  $h_j$ . Rational quadratic data give polynomial bit complexity. Every policy uses one fixed book, preserves the supplied targets exactly, commits intermediate service before its lottery, and respects every realization ceiling.*

*Proof.* Take a feasible ordered book. Every target belongs to exactly one interval  $[u, v)$  between consecutive levels, or to the final interval  $[u, \infty)$ . In the former case  $u \leq t_j \leq b_j \leq \tau_j$ , so the lower endpoint is eligible. If  $v$  is eligible, the canonical response is its chord at  $t_j$ . If  $v$  is ineligible, every later level is ineligible, and the highest eligible book level is  $u$ ; the response is the second line of (5). A final-interval branch has the response in (6). These intervals partition the branches, including targets equal to a selected level. The book’s operating value is therefore the sum of its edge and stopping rewards.

The recurrence enumerates ordered paths with exactly  $\ell$  vertices. It pays every charge once, with the first charge outside the recurrence. Conversely, each such path starting no higher than the smallest target defines a feasible book, and the preceding partition computes its exact value. Maximization proves both directions of the claimed equality. Taking the best exact cardinality gives an at-most budget and does not assume that an unused additional symbol should be purchased. There are  $O(N^2)$  edges, each computed by scanning the branches, and  $O(MN^2)$  transitions. Each rational path value is a sum of polynomially many input-derived rational terms, so ordinary rational arithmetic and comparison have polynomial bit complexity.  $\square$

**Corollary 3.2** (Global catalog design at saturation). *If  $B = \bar{b}$ , then  $P_s(b) = \mathcal{V}_{s,\mathcal{A}}^\tau(\bar{b}; \rho)$  for every  $s$ . Hence joint catalog design at a saturated promise, including arbitrary nonnegative opening charges and branch-specific realization ceilings, has the complexity in Theorem 3.1, with both  $k$  and  $s$  variable.*

*Proof.* Positive weights and  $t_j \leq b_j$  imply  $\sum_j \pi_j t_j = \bar{b}$  only when every  $t_j = b_j$ . There is no remaining target optimization. Apply the theorem.  $\square$

This corollary is a same-budget global result, not a priced relaxation. At an interior promise, the theorem optimizes the entire alphabet conditional on  $t$ ; it does not certify that the selected target vector is jointly optimal. Section EC.12 resolves that target choice at fixed branch count. The faster saturated Monge construction in Appendix D concerns its stronger structural assumptions and is retained rather than replaced.

### 3.1 Compression, opening costs, and symbol prices

Take any candidate pool  $U$ , including a union of supported books. Applying the theorem with  $\mathcal{A} = U$  gives the optimal sub-book at every allowed cardinality, not just a sequence of greedy deletions. Applying it on the whole catalog can only improve the conditional value. No nestedness of optimal books is required. In particular, comparing budgets  $m, m + 1, \dots, |U|$  gives the complete conditional memory–value frontier and identifies the value purchased by each additional symbol.

If an additional uniform certification fee  $\kappa \geq 0$  is imposed, the exact conditional value is

$$\max_{1 \leq \ell \leq M} \{F_\ell(t) - \kappa \ell\}. \quad (8)$$

The upper envelope has finitely many breakpoints, obtained from the computed exact-cardinality values. Chosen cardinalities may be tied, but a largest optimal cardinality is nonincreasing in  $\kappa$ . Indeed, optimality at  $\kappa_1 < \kappa_2$  and cardinalities  $\ell_1, \ell_2$  gives  $(\kappa_2 - \kappa_1)(\ell_1 - \ell_2) \geq 0$ . This statement concerns cardinality, not nesting of the selected levels. General nonuniform charges are already included in the recurrence and do not require the uniform-fee interpretation.

The structural advantage over a raw union is that charges enter the optimization itself. A costly level may be discarded even when capacity remains. Nevertheless, an optimal conditional compression can be inferior to a policy with different branch targets. Our returned operational policy therefore compares valid candidates at the *same* budget and keeps the best, including a supplied incumbent. This safeguard never converts a heuristic target portfolio into a global optimality theorem.

## 4 Response Types and the Complexity Map

Write  $P_m(B) = \mathcal{V}_{m, \mathcal{A}}^\tau(B; \rho)$  and  $R(c) = \sum_{a_i \in c} \rho_i$ . All results in this section concern a fixed finite catalog  $\mathcal{A} = (a_1 < \dots < a_N)$ . A response type consists of a cap, a shortfall cost function, and an eligible catalog prefix. Let  $d$  denote the number of such types; this is distinct from the quadratic reward-curvature parameter  $q$  in the continuous-design appendix.

**Proposition 4.1** (Exact response-type reduction). *Partition branches into groups with the same  $b_j$ , the same  $h_j$  on  $[0, b_j]$ , and the same set  $\mathcal{A} \cap [0, \tau_j]$ . Replace each group by one branch whose weight is the sum of its weights and whose ceiling is the minimum of its ceilings. The reduced and original joint catalog problems have the same optimum at every feasible  $B$  and every budget  $m$ . A reduced solution lifts by assigning its target and canonical lottery to every branch in its group.*

*Proof.* Every book is a subset of  $\mathcal{A}$ . Group members therefore have exactly the same eligible book and the same concave response  $v_c$ . For a group of total weight  $w$ , replace its targets by their weighted average  $\bar{t}$ . This preserves its contribution to the root equality, keeps the average between the common anchor and cap, and gives  $wv_c(\bar{t}) \geq \sum_{j \text{ in group}} \pi_j v_c(t_j)$ . Neither the book nor its charge changes. Conversely, the common reduced target and its canonical realization satisfy every original ceiling because the eligible catalog prefix is unchanged. The two feasible mappings prove equality of optimal values.  $\square$

The common-ceiling homogeneous aggregation argument already appears in the retained continuous analysis. The present statement uses catalog-prefix equivalence and permits distinct numerical ceilings. Equal caps alone do not suffice when either the cost or eligible prefix differs. Forming rational quadratic signatures by binary searching the ordered catalog takes  $O(k \log N)$  prefix comparisons, plus grouping and output work. With hashing, grouping uses expected linear dictionary operations; comparison-based sorting instead adds  $O(k \log k)$  comparisons. General cost-function equality requires a supplied exact representation or a verified type partition.

**Proposition 4.2** (Positive complexity regimes). *For rational quadratic primitives: (i) one response type or  $B = \bar{b}$  is solved by one invocation of Theorem 3.1; (ii) for every fixed  $m$ , exact joint catalog design is polynomial in  $N$ ,  $k$ , and input bit length by enumerating at most  $\sum_{s=1}^m \binom{N}{s}$  books and applying the fixed-book allocator; and (iii) the adaptive method below has polynomial work per price per node and a finite additive scheme in dimension  $d - 1$ .*

*Proof.* With one type the root equality fixes its target. At saturation positive weights force every target to its cap. For a specified book the concave allocation and its supporting-price certificate are polynomial for rational quadratic data, so enumerating the stated finite collection proves (ii). Theorems 5.1 and 6.1 prove (iii).  $\square$

Enumeration is an XP bound in  $m$ , not a fixed-parameter bound of the form  $F(m)N^c$  with a uniform exponent. A hardness assertion with a fixed small  $m$  would contradict (ii) unless P equals NP. We establish neither general NP-hardness nor a polynomial-time algorithm when both  $m$  and the number of distinct types vary. The additive scheme below remains exponential in  $d$  in its worst case. This map distinguishes proved upper bounds from unresolved lower bounds rather than treating the earlier grid's dimension as an intrinsic impossibility.

## 5 An Exact Two-Sided Price Oracle

After any exact type reduction, retain the index  $j$  for a branch and its weight  $\pi_j$ . A target box  $X = [\ell, u]$  satisfies  $0 \leq \ell_j \leq u_j \leq b_j$  and intersects  $\sum_j \pi_j t_j = B$ . Only anchors in

$$I_X = \left\{ i : a_i \leq \min_j u_j, \quad \sum_j \pi_j \max\{\ell_j, a_i\} \leq B \right\} \quad (9)$$

can meet the equality inside this box. Because  $B \leq \sum_j \pi_j u_j$ , these conditions are also sufficient for a fixed book's box feasibility. They depend on its lowest level, not on its remaining levels.

Let  $g(x) = f(x) - \lambda x$  at a fixed real price  $\lambda$ . For a book whose first index lies in  $I_X$ , relax only the

aggregate equality and define

$$S_X(\lambda) = \max_{\substack{\emptyset \neq c \subseteq \mathcal{A}, |c| \leq m \\ c_1 = a_i, i \in I_X}} \left\{ \sum_j \pi_j \max_{\max\{\ell_j, c_1\} \leq t \leq u_j} [v_{j,c}(t) - \lambda t] - R(c) \right\}, \quad (10)$$

$$U_X(\lambda) = \lambda B + S_X(\lambda). \quad (11)$$

Every  $U_X(\lambda)$  bounds the best original-budget joint design in  $X$ . No zero-duality-gap assumption is made. For a general cost define the clipped scalar support

$$\Phi_j(s, t; \lambda) = \max_{s \leq y \leq t} \{-h_j(y) - \lambda y\}.$$

For quadratic costs it is obtained by clipping  $-\lambda/\gamma_j$  to  $[s, t]$  when  $\gamma_j > 0$ ; for zero curvature choose the endpoint with the better linear value. Write  $\psi_j(z) = \Phi_j(\max\{0, \ell_j - z\}, u_j - z; \lambda)$  whenever  $z \leq u_j$ .

### 5.1 The increasing and decreasing sides

For  $x < z$  define an upper-crossing edge and its terminal value by

$$A^+(x, z) = \sum_{j: x \leq u_j < z} \pi_j \begin{cases} g(x) + \frac{u_j - x}{z - x} [g(z) - g(x)], & z \leq \tau_j, \\ g(x) + \psi_j(x), & z > \tau_j, \end{cases} \quad (12)$$

$$T^+(x) = \sum_{j: u_j \geq x} \pi_j [g(x) + \psi_j(x)]. \quad (13)$$

When  $\lambda > 0$  also define a lower-crossing edge, a lower terminal value, and a peak value:

$$A^-(x, z) = \sum_{j: x \leq \ell_j < z} \pi_j \begin{cases} g(x) + \frac{\ell_j - x}{z - x} [g(z) - g(x)], & z \leq \tau_j, \\ g(x) - h_j(\ell_j - x) - \lambda(\ell_j - x), & z > \tau_j, \end{cases} \quad (14)$$

$$T^-(x) = \sum_{j: \ell_j \geq x} \pi_j [g(x) - h_j(\ell_j - x) - \lambda(\ell_j - x)], \quad (15)$$

$$H(x) = \sum_{j: \ell_j < x \leq u_j} \pi_j g(x). \quad (16)$$

Only increasing-score arcs use  $A^+$ , and only decreasing-score arcs use  $A^-$ . Equal-score arcs may be assigned to either side. Set missing forward states to  $-\infty$ , and compute

$$\alpha_1(i) = \begin{cases} -\rho_i, & i \in I_X, \\ -\infty, & \text{otherwise,} \end{cases} \quad (17)$$

$$\alpha_r(v) = -\rho_v + \max_{\substack{i < v \\ g(a_i) \leq g(a_v)}} [\alpha_{r-1}(i) + A^+(a_i, a_v)], \quad 2 \leq r \leq m.$$

This table pays the charge for its last level. For positive prices a backward table excludes the charge for its first level:

$$\begin{aligned} \beta_1(i) &= T^-(a_i), \\ \beta_s(i) &= \max \left\{ T^-(a_i), \max_{\substack{v > i \\ g(a_v) \leq g(a_i)}} [A^-(a_i, a_v) - \rho_v + \beta_{s-1}(v)] \right\}, \quad 2 \leq s \leq m. \end{aligned} \quad (18)$$

**Theorem 5.1** (Two-sided box-price decomposition). *If  $I_X$  is nonempty, the exact support in (10) is*

$$S_X(\lambda) = \begin{cases} \max_{1 \leq r \leq m, i} \{\alpha_r(i) + T^+(a_i)\}, & \lambda \leq 0, \\ \max_{1 \leq r \leq m, i} \{\alpha_r(i) + H(a_i) + \beta_{m-r+1}(i)\}, & \lambda > 0. \end{cases} \quad (19)$$

*The maximizing book and scalar target supports are recoverable. Computing one price bound takes  $O((d+m)N^2)$  arithmetic operations and  $O(N^2 + mN)$  storage, where  $d$  is the number of branches after type reduction. General costs additionally require  $O(dN^2)$  clipped scalar-oracle calls. Rational quadratic data and a rational price give an exact rational bound in polynomial bit complexity for each node.*

*Proof.* Concavity of  $g$  implies that its values on any ordered book increase to a peak and then decrease; a tie can be placed on either side. For  $\lambda \leq 0$ , strict increase of  $f$  makes all scores strictly increasing, so the peak is the final level. Consider first a positive price and a chosen peak  $p$ .

If  $u_j < p$ , exactly one increasing edge  $[x, z)$  crosses  $u_j$ . If  $z$  is eligible, the interpolant of  $g$  is nondecreasing through this edge and is maximized on the permitted target interval at  $u_j$ . No later segment below  $u_j$  exists. If  $z$  is ineligible, every later book level is ineligible, so  $x$  is the last usable level and clipped shortfall maximization gives  $g(x) + \psi_j(x)$ . This proves the upper-crossing formula. Its clipping is essential when  $\ell_j > x$ .

If  $\ell_j < p \leq u_j$ , the peak is eligible because  $p \leq u_j \leq b_j \leq \tau_j$ . It attains  $g(p)$  at zero shortfall. All eligible chord values are at most  $g(p)$ , and positive-price shortfall cannot improve them. This proves the middle term.

If  $\ell_j \geq p$ , the interpolant on the decreasing side is nonincreasing. Each shortfall tail is also nonincreasing because  $h_j$  is nondecreasing and  $\lambda > 0$ . The smallest permissible target,  $\ell_j$ , therefore maximizes the branch's priced payoff. Its crossing edge gives (14); when the next level is ineligible it uses the singleton plus the required shortfall. Targets beyond the last level give (15). Equality at the peak belongs to this third group, preventing double counting.

These three groups partition the branches. Upper crossings before the peak, the peak, and lower crossings after the peak thus decompose the exact priced value of every book, not merely a dominant subset of books. Forward states enumerate all increasing prefixes starting at a feasible anchor. Backward states enumerate all decreasing suffixes, with an optional stop. Gluing a prefix of  $r$  levels to a suffix of at most  $m - r + 1$  levels counts the shared peak once, counts its charge once, and uses at most  $m$  levels. Every feasible ordered book has such a representation, and every represented path is a feasible book for the box. This proves the positive-price formula.

For nonpositive prices, the best point on an eligible increasing chord up to the cap is the cap. Where the next level is ineligible, the last singleton plus clipped shortfall gives the scalar optimum; the same argument gives the final tail. Hence (12) and (13) partition all branches, proving the first case.

The edge construction scans at most  $d$  branches for each ordered pair. Each table has  $mN$  states and at most  $N$  transitions per state. Backpointers reconstruct the book, and scalar maximization reconstructs its supports. For quadratic rational inputs each edge uses a polynomial number of rational operations; each path sums polynomially many terms. Their numerator and denominator lengths remain polynomial in the input, price, and box-endpoint bit lengths. This is a per-node bit-complexity statement, not a bound independent of search depth.  $\square$

**Why a second side is necessary.** Take equal weights, targets and caps  $(1/5, 1/2, 4/5)$ , unit shortfall curvature, unit ceilings, catalog  $\{1/5, 1/2, 4/5\}$ , zero charges, budget three, and reward  $f(x) = 2x - x^2/2$ . At price  $3/2$ , the complete book has scores  $(2/25, 1/8, 2/25)$ . On the singleton target box its exact joint value is  $169/200$ . Restricting to nondecreasing-score books gives only  $83/100$ , a shortfall of  $3/200$ . Thus the unrestricted deletion argument in Theorem A.1 cannot be reused on a target box. The new recurrence repairs that structural obstruction rather than placing the old one-sided oracle inside a generic partitioning procedure.

## 6 Adaptive Joint Design and Verifiable Global Intervals

Let  $\Theta = \{t \in [a_1, b] : \sum_j \pi_j t_j = B\}$ . A box can be reduced without losing any point of its intersection with  $\Theta$ . With  $s_\ell = \sum_j \pi_j \ell_j$  and  $s_u = \sum_j \pi_j u_j$ , its exact coordinate projections are

$$\ell'_j = \max \left\{ \ell_j, \frac{B - s_u + \pi_j u_j}{\pi_j} \right\}, \quad u'_j = \min \left\{ u_j, \frac{B - s_\ell + \pi_j \ell_j}{\pi_j} \right\}. \quad (20)$$

The box is discarded only if  $B \notin [s_\ell, s_u]$  or an interval is empty. Formula (20) follows by allocating all other coordinates to their upper or lower endpoints; continuous interval filling proves attainability of each projected endpoint.

At each retained box compute  $U_X = \min_{\lambda \in \Lambda} U_X(\lambda)$  for any finite price set containing zero. Every recovered book has an anchor in  $I_X$  and hence has a feasible allocation satisfying the original equality. Optimize that book's targets exactly, either within  $X$  or over the entire original target region, and update the feasible incumbent  $L$ . The implementation uses global fixed-book allocation and caches repeated books. This changes neither the original budget nor the charge schedule. Root price bisection is a choice of bound-strengthening prices, not an assertion of zero duality gap or a prerequisite for validity.

Retain the leaves of a binary cover of  $\Theta$ , select a leaf with largest upper bound, and bisect one independent target coordinate at its midpoint. Apply (20) to both children. Their intersections with  $\Theta$  cover their parent's intersection; the common boundary may overlap. A dependent coordinate  $j_0$  is recovered through the equality. Use as splitting coordinate the longest weighted side  $\pi_j(u_j - \ell_j)$  among  $j \neq j_0$ . The code chooses a largest-weight coordinate as  $j_0$ ; this choice is not needed for validity.

**Theorem 6.1** (Adaptive original-budget certificate and finite accuracy). *Assume  $f$  and all  $h_j$  are Lipschitz on their relevant compact intervals and put  $L_v = \max\{L_f, \max_j L_{h,j}\}$ . Use the exact oracle in Theorem 5.1, include price zero at every node, and evaluate the returned zero-price book by a feasible exact fixed-book allocation. At every interruption,*

$$L \leq P_m(B) \leq U = \max\{L, \max_{X \text{ a leaf}} U_X\}. \quad (21)$$

*For every  $\varepsilon > 0$ , best-upper-bound selection and longest-weighted-side bisection terminate with  $U - L \leq \varepsilon$ . If  $d = 1$  the root is a singleton and is exact. For fixed  $d > 1$ , a conservative full-tree bound is  $2^{H+1} - 1$  nodes, where*

$$H = \sum_{j \neq j_0} \max \left\{ 0, \left\lceil \log_2 \frac{2L_v(d-1)w_j^0}{\varepsilon} \right\rceil \right\}, \quad w_j^0 = \pi_j(u_j^0 - \ell_j^0). \quad (22)$$

*A zero width contributes zero. With  $L_v = 0$ , one root evaluation suffices. The bound is polynomial in numerical  $1/\varepsilon$  for fixed  $d$ , and exponential in  $d$  in general; adaptive pruning can finish before this worst-case cover is constructed.*

*Proof.* A feasible policy supplies the lower bound. Weak duality and the complete leaf cover supply the upper bound, irrespective of the search order or the quality of the incumbent. Thus pruning is valid whenever a leaf bound is at most  $L + \varepsilon$ , and retaining such leaves in a certificate is harmless.

For local tightness, let  $c^0$  maximize  $S_X(0)$  and let  $\hat{t}$  be its independently maximizing box targets. Condition (9) gives some equality-feasible  $t^0$  for that same book in  $X$ . The canonical response is continuous and piecewise concave with slopes bounded in absolute value by  $L_v$ : its chord slopes are bounded by  $L_f$  and its shortfall slopes by  $L_{h,j}$ . Charges cancel for the same book. Therefore

$$U_X \leq U_X(0) \leq \sum_j \pi_j v_{j,c^0}(t_j^0) - R(c^0) + L_v \sum_j \pi_j (u_j - \ell_j) \leq L + L_v \sum_j \pi_j (u_j - \ell_j). \quad (23)$$

An exact global fixed-book allocation can only improve the displayed feasible value. The inequality does not require a uniform bound on a recovery-compression penalty and does not mix two books.

After coordinate projection, the dependent weighted width is no larger than the sum of the other weighted widths. To see this, choose equality-feasible points attaining its lower and upper endpoints and subtract their equalities. Thus the last sum in (23) is at most twice the sum of the independent weighted widths. No node needs further splitting once each independent width is at most  $\varepsilon/[2L_v(d-1)]$ . Before this occurs, a longest-width split halves a width above that threshold; projection cannot increase any width. Along a path, coordinate  $j$  can be split at most its summand in (22) before falling below the threshold. Hence every path needing refinement has at most  $H$  splits. The complete binary tree bound follows. At a singleton box the same calculation gives equality of the box upper bound and the fixed-target optimum. This also proves the one-type case.  $\square$

The result replaces completion of a full target net by certificate-directed refinement. It does not promise that refinement is inexpensive for every input. It supplies a structural polynomial node oracle and a

meaningful interval when it is not. An absolute tolerance is measured in payoff units. Our quadratic experiments normalize target levels to  $[0, 1]$  and use  $r = 2, q = 1$ ; they report both the absolute gap and its ratio to  $r + \max_j \gamma_j/2 + \sum_{i=1}^m \rho_{(i)}$ , where the charges are sorted decreasingly. This positive declared scale remains meaningful when net value is close to zero. Rescaling all rewards and charges requires rescaling the requested tolerance.

## 6.1 Independent checking and physical implementation

A certificate contains the original inputs, the exact type partition, an original-space lottery, every binary split, the authoritative leaves, and one upper-bounding price table per node. It need not contain every price tried. The checker verifies anchor bounds, every reachable forward Bellman inequality, every backward stopping and transition inequality, and the final maximum in (19). Upper tables need not equal the optimizer's tables; satisfying these inequalities is enough to dominate every represented path. It separately reconstructs both projected children at every split and verifies that no leaf is missing. Finally, it checks probabilities, selected charges, pre-draw service, the exact root promise, and supportwise realization limits without using the canonical response formula. The certificate therefore certifies all target regions, not just the sampled incumbents. Its storage is  $O(TmN + Td)$  rational entries for  $T$  recorded nodes, apart from the policy and input data; the checker recomputes edge values and has polynomial work per recorded node.

The terminal alphabet is only one physical resource. A type-reduced implementation additionally stores a history-to-type encoder, a target and lottery table, command constants, and probability representations. Arbitrary rational probabilities may require variable-length random-bit generation or a separate finite-precision error analysis; we do not count them as free writable memory. Offline certificates have the size just stated and are not assumed to fit in the terminal gateway. The total deployment cost is consequently not identified with  $\lceil \log_2 m \rceil$  bits. This separation is essential to interpreting the repeated-type computation as joint design of a restricted interface, rather than a claim about full-system memory or calibrated field performance.

## 7 Certified Response Coarsening for Distinct Histories

Exact type aggregation can leave one type per history. We now construct a controlled relaxation that also combines different caps and costs, and then recover a contract satisfying every original constraint. In this section the primitives are quadratic,

$$f(x) = rx - qx^2/2, \quad h_j(y) = \gamma_j y^2/2, \quad r \geq q \geq 0, \quad r > 0, \quad \gamma_j \geq 0,$$

with normalized levels in  $[0, 1]$ . The general canonical and continuous-design results retain their preceding assumptions. Write  $P$  for  $\mathcal{V}_{m, \mathcal{A}}^\tau(B; \rho)$  in this section. The catalog, charges, promise, and command budget are fixed throughout.

Partition the histories into groups  $G_1, \dots, G_d$ . A group must have the same *catalog eligibility prefix*:

$\mathcal{A} \cap [0, \tau_j]$  is identical for its members. Caps and shortfall curvatures need not agree. Define

$$W_g = \sum_{j \in G_g} \pi_j, \quad \bar{b}_g = W_g^{-1} \sum_{j \in G_g} \pi_j b_j, \quad \underline{\gamma}_g = \min_{j \in G_g} \gamma_j, \quad \bar{\tau}_g = \max_{j \in G_g} \tau_j. \quad (24)$$

The reduced model uses weight  $W_g$ , cap  $\bar{b}_g$ , cost curvature  $\underline{\gamma}_g$ , and ceiling  $\bar{\tau}_g$ . It preserves total participation capacity exactly. Require the *anchor guard*

$$\min_g \bar{b}_g = \min_j b_j. \quad (25)$$

Isolating one minimum-cap history suffices and adds at most one group. This restriction is essential: averaging caps 1/5 and 4/5 with equal weights would otherwise admit the one-command book  $\{1/2\}$  at promise 1/2, although it violates the first original cap. Let  $\bar{P}_G$  denote the reduced joint optimum. The reduced model is an optimistic bound, not a replacement of the contract's participation constraints.

For a feasible book  $c$ , let  $d_g(c)$  be its largest command in the common eligible prefix of group  $g$ . Define the representative response  $\bar{v}_{g,c}$  by the canonical formula with curvature  $\underline{\gamma}_g$ . We use its mathematical extension to targets up to  $\max_{j \in G_g} b_j$  when comparing individual histories. This extension is concave and has Lipschitz constant

$$L_g = \max\{r, \underline{\gamma}_g \max_{j \in G_g} b_j\}. \quad (26)$$

The extension does not enlarge the reduced optimization's cap. With  $a_1$  the lowest catalog level, define an explicit, book-independent error budget

$$\Delta_G = \sum_g \left[ 2L_g \sum_{j \in G_g} \pi_j (\bar{b}_g - b_j)_+ + \frac{1}{2} \sum_{j \in G_g} \pi_j (\gamma_j - \underline{\gamma}_g) (b_j - a_1)^2 \right]. \quad (27)$$

**Theorem 7.1** (Optimistic coarsening and original-contract recovery). *Under (25) and common catalog eligibility within each group,*

$$P \leq \bar{P}_G \leq P + \Delta_G. \quad (28)$$

*More specifically, consider any feasible reduced policy with book  $c$ , targets  $s_g$ , and net value  $\bar{V}$ . There exists a constructive lift with the same book, original targets  $t_j \leq b_j$ , the exact original promise, and every original realization ceiling. Its value  $V$  satisfies*

$$\bar{V} - V \leq \Delta_G(s, t, c) \leq \Delta_G, \quad (29)$$

$$\Delta_G(s, t, c) = \sum_g \left[ L_g \sum_{j \in G_g} \pi_j |t_j - s_g| + \frac{1}{2} \sum_{j \in G_g} \pi_j (\gamma_j - \underline{\gamma}_g) (t_j - d_g(c))_+^2 \right].$$

*If a reduced solver supplies a feasible  $\bar{V}$  and certified upper bound  $\bar{U}$ , then*

$$V \leq P \leq \bar{U}, \quad \bar{U} - V \leq (\bar{U} - \bar{V}) + \Delta_G(s, t, c) \leq (\bar{U} - \bar{V}) + \Delta_G. \quad (30)$$

*No command augmentation or extra opening charge is required.*

*Proof.* The feasible anchors of the two models coincide by (25); both have the same total cap. For any original feasible  $(c, t)$ , set  $s_g = W_g^{-1} \sum_{j \in G_g} \pi_j t_j$ . These means satisfy the reduced caps and root equality. Because lowering the cost curvature increases each response, concavity gives

$$\sum_{j \in G_g} \pi_j v_{j,c}(t_j) \leq \sum_{j \in G_g} \pi_j \bar{v}_{g,c}(t_j) \leq W_g \bar{v}_{g,c}(s_g).$$

Summing and subtracting the unchanged charge proves  $P \leq \bar{P}_G$ .

To lift a reduced policy, first set  $z_j = \min\{s_g, b_j\}$  for  $j \in G_g$ . The group deficit is  $D_g = W_g s_g - \sum_{j \in G_g} \pi_j z_j = \sum_{j \in G_g} \pi_j (s_g - b_j)_+$ . Fill this deficit in any fixed order by increasing  $z_j$  up to  $b_j$ . Sufficient capacity exists because  $s_g \leq \bar{b}_g$ . The resulting  $t$  preserves each group mean and therefore the root equality. Every target remains at least  $\min c$ : the anchor guard makes all original caps at least this anchor. A clipped history is already at its cap; only histories initially at  $s_g$  can be increased. Consequently

$$\sum_{j \in G_g} \pi_j |t_j - s_g| = 2D_g \leq 2 \sum_{j \in G_g} \pi_j (\bar{b}_g - b_j)_+. \quad (31)$$

Apply the canonical original policy at these targets. Below  $d_g(c)$  it uses a lottery entirely in the common eligible prefix and has zero intermediate service. Above  $d_g(c)$  it uses that singleton command and intermediate service  $t_j - d_g(c)$ , so realized total equals  $t_j \leq b_j \leq \tau_j$ . Thus equal eligibility, rather than equality of numerical ceilings, is exactly what the lift requires.

On the shared target domain,  $\bar{v}_{g,c}(t) - v_{j,c}(t) = (\gamma_j - \underline{\gamma}_g)(t - d_g(c))_+^2/2$ . The chord slopes of  $f$  lie between zero and  $r$ , and the magnitude of each representative tail slope is at most  $\underline{\gamma}_g \max_{j \in G_g} b_j$ . Using (26) and (31) therefore proves (29); the cost term is bounded by  $(b_j - a_1)^2$ . Charges cancel. Apply the lift to a reduced optimum to prove the second inequality in (28). Weak duality for the reduced model and the same loss calculation prove (30).  $\square$

The reported certificate uses the measured interval  $[V, \bar{U}]$ , not just the worst-case sum in (30). It may be informative even when  $\Delta_G$  is large. Exact reallocation for the recovered book on the original histories can improve  $V$  without affecting the upper bound. The checker independently recomputes the representative model, anchor guard, error budgets, and group mass identities, verifies the inner price-path and cover certificate, and checks original lottery probabilities, intermediate service, charges, and all realized ceilings. It never treats a successful reduced solve as a successful original solve merely because the reduced stopping tolerance was met.

## 7.1 A variable-history additive approximation scheme

Let  $E$  be the number of distinct eligible catalog prefixes and  $\Gamma = \max_j \gamma_j$ . Set  $L = \max\{r, \Gamma\}$ . Partition first by eligibility, then by intervals of width  $\beta > 0$  for caps and  $\eta > 0$  for cost curvatures; isolate a minimum-cap history as the anchor guard. The remaining groups can contain arbitrarily many distinct caps and curvatures.

**Lemma 7.2** (Dispersion bound). *For this partition,*

$$\Delta_G \leq L\beta/2 + \eta/2, \quad d \leq 1 + E(1 + \lceil 1/\beta \rceil)(1 + \lceil \Gamma/\eta \rceil). \quad (32)$$

*Proof.* For a random variable  $X$  in an interval  $[a, a + \beta]$  with mean  $\mu$ , convexity of  $(\mu - X)_+$  bounds its expectation by the chord between the endpoints. This gives  $\mathbb{E}(\mu - X)_+ \leq (\mu - a)(a + \beta - \mu)/\beta \leq \beta/4$ . Apply this inequality to each group's weighted empirical caps. Its first contribution to (27) is at most  $LW_g\beta/2$ . Cost spread is at most  $\eta$ , and  $0 \leq b_j - a_1 \leq 1$ , so its second contribution is at most  $W_g\eta/2$ . Sum the group weights. The bin counts on the bounded cap and curvature intervals, multiplied by the eligibility count and augmented by the singleton guard, prove the dimension bound.  $\square$

**Theorem 7.3** (Additive scheme with variable history count). *For rational quadratic input and any  $\varepsilon > 0$ , use  $\beta = \varepsilon/(2L)$  and  $\eta = \varepsilon/2$  and solve the reduced joint problem to additive tolerance  $\varepsilon/2$  by Theorem 6.1. The lifted policy is feasible for the original problem and has value at least  $P - \varepsilon$ . The reduced dimension is bounded by*

$$d_\varepsilon = 1 + E(1 + \lceil 2L/\varepsilon \rceil)(1 + \lceil 2\Gamma/\varepsilon \rceil), \quad (33)$$

*independently of the number  $k$  of histories. For fixed  $E$ , bounded  $r$  and  $\Gamma$ , and fixed positive accuracy, this is a polynomial-time additive scheme in the variable history count, catalog size, command budget, and rational input length. Its accuracy dependence is not asserted to be polynomial.*

*Proof.* Lemma 7.2 gives  $\Delta_G \leq \varepsilon/2$ . Theorem 7.1 then yields a global original-space gap at most  $\varepsilon$ . To make the complexity claim explicit, for  $d > 1$  a conservative reduced full-tree bound is  $2^{H+1} - 1$ , with

$$H = (d - 1) \max\{0, \lceil \log_2(4L(d - 1)/\varepsilon) \rceil\}.$$

Indeed, every initial weighted side is at most one, the reduced Lipschitz constant is at most  $L$ , and the inner tolerance is  $\varepsilon/2$  in (22). One price-zero oracle per node already ensures this bound; finitely many additional root prices can strengthen it. Each quadratic price oracle takes  $O((d + m)N^2)$  rational arithmetic operations, and exact fixed-book allocation is polynomial. Group construction and original-space policy recovery are polynomial in  $k$  and  $N$ . The constructed rational means, quadratic stationary points, and dyadic subdivisions have bit length polynomial in the input length for fixed  $d$  and  $H$ . Substitution of (33) proves the stated parameterized dependence. The single-group case requires only the root.  $\square$

Common realization ceilings give  $E = 1$  even as both the catalog and the number of distinct histories grow. More generally, a bounded number of distinct ceilings bounds  $E$  independently of catalog size. When eligibility complexity or curvature scale grows without restriction, (33) exposes that cost rather than hiding it in  $k$ . The conservative bins can still leave every history separate at a tight requested tolerance; the dimension bound is not a claim of a small constant in practice. Coarser partitions are accepted only through the measured original-space interval. The result is neither a multiplicative approximation for potentially negative net payoffs nor a fully polynomial approximation scheme. Additive tolerances must be rescaled

together with rewards, costs, and opening charges. The finite-catalog restriction remains explicit; the separate continuous-design results do not inherit this accuracy claim automatically.

## 7.2 Refinement and the distinction between bounds and decisions

**Proposition 7.4** (Monotone optimistic refinement). *If  $G'$  refines  $G$ , both partitions respect eligibility, and both satisfy the anchor guard, then*

$$P \leq \bar{P}_{G'} \leq \bar{P}_G. \quad (34)$$

*The largest original-space feasible value and the smallest certified upper bound obtained across any sequence of such reduced solves form a nested valid interval. Repeated splitting reaches exact response groups or singletons after finitely many splits. At that point the coarsening error is zero, and a completed positive-accuracy inner solve gives the same original-space tolerance.*

*Proof.* Apply the first part of Theorem 7.1 to the refined model, grouping its subgroups into the coarse groups. Weights and weighted caps compose, minima of subgroup costs equal coarse minima, and eligibility is unchanged. Every original policy is a lower witness and every reduced upper bound is an upper witness, so taking their extrema is valid. A nontrivial split increases the number of groups; at most  $k - d$  such splits are needed to reach singletons. Exact response groups have constant caps and curvatures, making both terms in (27) zero.  $\square$

This proposition concerns reduced *optima*, not a false monotonicity claim for separately interrupted upper bounds. Our implementation carries the minimum upper and maximum lower from all stages. It splits a positive-error group, favoring the cap or curvature coordinate with the larger error contribution. That choice affects speed, not validity. Offline grouping also does not make the history encoder, individual cap table, lifted probabilities, or certificates free memory. The final policy and its implementation requirements remain indexed by the original histories.

## 8 Harmonic Pooling and Square-Root Cost Coarsening

Minimum-curvature aggregation discards how much service the low-cost histories can absorb. We strengthen its upper model without increasing the number of target coordinates. Retain the quadratic primitives, eligibility groups, and anchor guard of Section 7. For group  $g$ , write  $w_j = \pi_j/W_g$  and define

$$A_g = \sum_{j \in G_g} w_j \gamma_j, \quad H_g = \begin{cases} (\sum_{j \in G_g} w_j / \gamma_j)^{-1}, & \min_{j \in G_g} \gamma_j > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (35)$$

The harmonic representative has the same weighted cap, weight, and eligible prefix as before, but curvature  $H_g$ . Let  $P_G^H$  and  $P_G^{\min}$  be the harmonic and minimum-curvature joint optima. Their feasible books, opening charges, and target constraints are identical. Harmonic pooling and the inequalities between means are classical (Liao and Wu, 2015); the issue is whether they remain valid under a shared menu, exact root obligation, cap clipping, and supportwise realization limits.

**Theorem 8.1** (Dominating harmonic relaxation). *For every admissible partition,*

$$P \leq P_G^H \leq P_G^{\min}. \quad (36)$$

*For a fixed book, every original group allocation of mean  $s$  has operating value at most  $W_g v_{g,c}^H(s)$ . Moreover,  $H_g$  is the largest constant  $H$  such that  $\sum_j w_j \gamma_j y_j^2 \geq H(\sum_j w_j y_j)^2$  for all nonnegative service vectors. If  $G'$  refines  $G$  and both satisfy the guard, then  $P \leq P_{G'}^H \leq P_G^H$ .*

*Proof.* Fix a group and book, with last eligible level  $d$ . Set  $x_j = \min(t_j, d)$  and  $y_j = (t_j - d)_+$ . The common interpolant  $F_c$  is increasing and concave on the eligible book. With bars denoting weighted means, Jensen's and Cauchy–Schwarz inequalities give

$$\sum_j w_j \{F_c(x_j) - \gamma_j y_j^2/2\} \leq F_c(\bar{x}) - H_g \bar{y}^2/2 \leq F_c(\min(s, d)) - H_g (s - d)_+^2/2.$$

Indeed,  $\bar{x} + \bar{y} = s$ ,  $\bar{x} \leq d$ , and reward increases while cost decreases as  $\bar{x}$  increases. If a curvature is zero, the cost inequality is simply nonnegativity. The weighted means respect the reduced caps and root equality, and the guard preserves feasible anchors. Summing and subtracting unchanged charges proves the first inequality in (36). Since  $H_g \geq \min_j \gamma_j$ , the harmonic response never exceeds the minimum-curvature response, proving the second.

For positive curvatures, equality in the cost inequality holds at  $y_j = H_g u / \gamma_j$  for any prescribed mean  $u \geq 0$ ; if a curvature is zero, put all service on zero-curvature histories. Thus no larger uniform constant works. This sharpness assertion does not remove individual cap constraints. Finally, harmonic aggregation composes: the reciprocal of the coarse curvature is the subgroup-weighted mean of reciprocal subgroup curvatures. A zero curvature propagates through either aggregation. Applying the first argument to the refined representatives proves monotone refinement.  $\square$

## 8.1 Optimal within-group recovery and an explicit loss bound

For a reduced policy  $(c, s)$ , first form the clipping lift  $t^0$  from Theorem 7.1. Next optimize each original group's allocation for the same book and the same group mean  $s_g$ , using the concave fixed-book allocator. This second step cannot reduce value and introduces neither new commands nor new charges. It is distinct from changing the group means or selecting another book. Each group supplies a scalar supporting price, which independently certifies its allocation by direct maximization over eligible knots, the cap, and tail stationary points.

Let  $b_g^+ = \max_{j \in G_g} b_j$ ,  $L_{gj} = \max\{r, \gamma_j b_j^+\}$ , and  $L_g^+ = \max_j L_{gj}$ . Define

$$\Delta_G^H = \sum_g \left[ 2L_g^+ \sum_{j \in G_g} \pi_j (\bar{b}_g - b_j)_+ + \frac{W_g}{2} (A_g - H_g) (\bar{b}_g - a_1)^2 \right], \quad (37)$$

$$\delta_G^H(c, s) = \sum_g \left[ \sum_{j \in G_g} \pi_j L_{gj} |t_j^0 - s_g| + \frac{W_g}{2} (A_g - H_g) (s_g - d_g(c))_+^2 \right]. \quad (38)$$

**Theorem 8.2** (Same-book optimal lift and error composition). *The within-group optimal lift has net value  $V$  and satisfies every original constraint. If the reduced policy has value  $\bar{V}$  and certified upper bound  $\bar{U}$ , then*

$$V \leq P \leq \bar{U}, \quad 0 \leq \bar{V} - V \leq \delta_G^H(c, s) \leq \Delta_G^H, \quad \bar{U} - V \leq (\bar{U} - \bar{V}) + \Delta_G^H. \quad (39)$$

*In particular,  $P_G^H \leq P + \Delta_G^H$ . For rational quadratic data, recovery and its supporting-price certificates take polynomial time in the original history count and input bit length.*

*Proof.* The clipping construction preserves the group means and individual caps, so each within-group allocation is feasible. Its canonical lotteries respect the original eligible prefix and realization ceilings. A supporting-price water-level sweep gives the exact concave optimum and cannot be worse than  $t^0$ . On the mathematical extension to  $[\min c, b_g^+]$ , each original response is  $L_{gj}$ -Lipschitz, and

$$W_g v_{g,c}^H(s_g) - \sum_{j \in G_g} \pi_j v_{j,c}(s_g) = \frac{W_g}{2} (A_g - H_g)(s_g - d_g(c))_+^2.$$

Subtract the clipping lift and apply the Lipschitz bounds. The clipping distance is  $2 \sum_j \pi_j (s_g - b_j)_+$ ; also  $s_g \leq \bar{b}_g$  and  $d_g(c) \geq a_1$ . These facts give (38) and (37). Theorem 8.1 applied to the optimal lift at its unchanged group means gives  $V \leq \bar{V}$ . Charges cancel throughout. Weak duality and then a reduced optimal policy prove the remaining assertions. The allocator uses rational breakpoints and scalar quadratic stationary points, with polynomial encoding length.  $\square$

The harmonic optimum is always tighter than the minimum-curvature optimum, but its displayed worst-case defect need not be smaller:  $L_g^+$  uses the largest original curvature. Likewise, independently interrupted searches need not produce ordered upper bounds. Retain the minimum certified upper and maximum original feasible lower across methods. This gives a nested valid interval without a false dominance claim about every runtime or recovered policy.

The pooling bound can be exact with heterogeneous costs. With one command  $d = 0$ , equal weights, caps 9/10, curvatures 1 and 4, and mean obligation 1/4,  $H_g = 8/5$ . The allocation  $(2/5, 1/10)$  is feasible and has value  $-1/20$ , exactly the harmonic value; the minimum-curvature model gives  $-1/32$ . In general, for positive curvatures and mean  $s \geq d$ , equality holds when  $t_j = d + H_g(s - d)/\gamma_j \leq b_j$  for every member. A tight cap or exhausted zero-cost capacity can prevent equality; such cases require refinement rather than a completion claim based only on the reduced model.

## 8.2 Square-root bins improve the accuracy-dependent dimension

The arithmetic–harmonic gap is second order in the spread of square-root curvatures. This avoids assuming a positive lower cost curvature.

**Lemma 8.3** (Square-root dispersion). *For nonnegative curvatures in  $[u, v]$  and positive normalized weights,*

$$0 \leq A - H \leq (\sqrt{v} - \sqrt{u})^2. \quad (40)$$

If every group has cap spread at most  $\beta$  and square-root curvature spread at most  $\eta$ , then, with  $L = \max\{r, \Gamma\}$  and  $\Gamma = \max_j \gamma_j$ ,

$$\Delta_G^H \leq L\beta/2 + \eta^2/2. \quad (41)$$

*Proof.* For  $0 < u \leq \gamma \leq v$ ,  $(\gamma - u)(\gamma - v) \leq 0$  implies  $\gamma + uv/\gamma \leq u + v$ . Taking means and writing  $T = \sum_j w_j/\gamma_j$  gives  $A - H \leq u + v - uvT - 1/T \leq u + v - 2\sqrt{uv}$ . When  $u = 0$ ,  $A - H \leq A \leq v$  proves the same bound, including zero curvatures. Nonnegativity follows from Cauchy–Schwarz. For caps, the bounded-mean deviation inequality in Lemma 7.2 gives  $\sum_j \pi_j (\bar{b}_g - b_j)_+ \leq W_g \beta/4$ . Apply (40),  $\bar{b}_g - a_1 \leq 1$ , and sum the weights in (37).  $\square$

**Theorem 8.4** (Square-root-coarsened variable-history scheme). *For rational quadratic finite-catalog inputs and  $\varepsilon > 0$ , partition by exact eligibility, cap bins of width  $\varepsilon/(2L)$ , and curvature intervals  $[\ell^2\varepsilon/2, (\ell + 1)^2\varepsilon/2]$ , for integers  $\ell \geq 0$ . Add the minimum-cap guard. Solve the harmonic reduced problem to tolerance  $\varepsilon/2$  and apply the optimal lift. The original policy is  $\varepsilon$ -optimal at the original command budget, with unchanged charges, root promise, and individual constraints. The number of reduced coordinates is at most*

$$d_\varepsilon^H = 1 + E(1 + \lceil 2L/\varepsilon \rceil)(1 + \lceil \sqrt{2\Gamma/\varepsilon} \rceil). \quad (42)$$

*For fixed eligibility complexity, bounded primitive scales, and fixed positive accuracy, running time is polynomial in the variable history count, catalog size, command budget, and rational input length.*

*Proof.* The squared endpoints ensure square-root spread at most  $\sqrt{\varepsilon/2}$ ; bin membership uses rational comparisons and integer square roots, not rounded floating boundaries. Lemma 8.3 gives defect at most  $\varepsilon/2$ , and Theorem 8.2 adds the inner tolerance. Counting cap and square-root bins proves (42). Use the finite adaptive-cover bound in Theorem 7.3, replacing its dimension by  $d_\varepsilon^H$ . Harmonic means of rational data and optimal lifts have polynomial bit length; grouping and recovery scan the original histories. This proves the claimed parameter regime.  $\square$

For fixed primitive scales, the worst-case dimension improves from order  $\varepsilon^{-2}$  to order  $\varepsilon^{-3/2}$ ; these are *dimension* bounds, not runtime exponents. The adaptive cover remains exponential in this dimension. Neither theorem is an FPTAS or a hardness classification for unrestricted eligibility complexity. The classical scalar inequalities are not claimed as new: the contribution is their integration into a dominating contract-preserving relaxation, an optimally allocated original-space lift, and the improved variable-history accuracy construction. All preceding minimum-curvature guarantees remain available, including at zero costs.

## 9 Joint Computational Evidence

The evidence now separates two questions. The preserved distinct-history study tests certified joint design against unaggregated boxes, priced-prefix search, all-book enumeration, and direct mixed-integer envelopes. Its 24 original-space successes at tolerance 0.001, including up to 256 distinct histories, remain in Tables 4–5. Those clustered-input results are not a universal runtime claim. The full protocol, four deliberately

unresolved coarse stress cases, their refinements, and the earlier R46 failures remain in the computational record and companion tables.

The new paired study isolates harmonic pooling under substantially dispersed costs. It contains 24 cases: twelve primary cases with four, six, or eight histories and twelve cases with 32, 128, or 256 histories. Catalog sizes are seven and nine, budgets range from one to three, and the four families vary common or dispersed caps, eligibility, saturated or interior promises, positive charges, and a zero-cost history. Within each shared eligibility prefix and cap bin, all curvatures enter one deliberately coarse group; both representatives use identical groups and a minimum-cap guard. Each receives 31 target boxes, four price refinements and reduced tolerance 0.0005. The requested original tolerance is 0.001.

Table 10 reports 14 harmonic certificates within tolerance, versus 11 for minimum-curvature aggregation. All 10 unresolved harmonic cases remain in the record; the largest gap is 0.0316. At the twelve primary all-book references, harmonic reduced optima are never larger and are strictly smaller in 5 cases. Nevertheless, a tighter relaxation need not improve every interrupted interval: cost pooling can leave cap dispersion dominant, and independently chosen price bounds can differ. We report both intervals and their valid intersection, not a fictitious pointwise runtime dominance.

All 48 paired upper/lower certificates pass independent semantic checking. The 24 direct tangent/secant MIP solves retain original support probabilities and service decisions; every solver status, time, node count, primal–dual gap, envelope error, and bracket width is reported in Table CR.2. These numerical intervals are not rational proofs. Exact regression checks comprise 32 ordered joint-optimum comparisons, 199 original-policy lifts, 65 dispersion checks, 96 accuracy partitions, 32 refinement comparisons, and 14 rejected semantic mutations. The study tests the new mathematical mechanism rather than replacing unresolved cases with selected successes. No operational population, charge calibration, or deployment is claimed from synthetic inputs.

## 10 Conclusion

The shared execution alphabet makes menu selection and service allocation inseparable. Eligibility-aware path bounds, adaptive target covers, and guarded history aggregation give original-budget joint certificates. Harmonic pooling strengthens the reduced upper model, optimal groupwise recovery restores the original contract, and square-root cost bins improve the accuracy-dependent dimension without excluding zero costs. The method preserves every installed-level charge, accepted obligation, participation cap, and realized ceiling.

The numerical evidence separates three objects: an optimistic reduced optimum, a feasible original contract, and their certified gap. The cost-heterogeneity ablation exhibits both tighter intervals and persistent cap-driven failures; neither is hidden by a reduced solver’s success. The continuous, institutional, Monge, compression, and inexact-support results retain their assumptions and proofs. The unresolved theoretical frontier is unrestricted eligibility complexity and practically small worst-case accuracy dependence, not feasibility recovery or original-budget certification. No empirical population calibration or deployment is inferred from the synthetic experiments.

## A Eligibility-Prefix Decomposition

The closed-cell formulation establishes exact continuous attainability. We now exploit the ordered interface to avoid enumerating either branch modes or catalog subsets in a price subproblem. The resulting bounds support finite global search and remain valid when that search stops early. This distinction matters: a price relaxation is not, without an additional equality, the original optimum.

### A.1 A reusable ordered-interface allocation class

Let  $\mathcal{A} = (a_1 < \dots < a_N) \subset [0, 1]$  be a catalog with nonnegative opening charges  $\rho_i$ . The common reward  $f$  is strictly increasing and concave. Branch  $j$  has weight  $\pi_j$ , target cap  $b_j$ , nondecreasing convex shortfall cost  $h_j$  with  $h_j(0) = 0$ , and an eligibility threshold  $\tau_j \geq b_j$ . Its response is (2) with eligible levels  $c \cap [0, \tau_j]$ . The renewal model has  $\tau_j = b_j + \delta$ ; the theorem below permits distinct thresholds without requiring a common overrun tolerance. This abstract class needs no acceptance or renewal interpretation. It describes any ordered shared interface with a common concave output reward, separable shortfall costs, and these target and support restrictions.

Write  $a_* = \min\{B, \min_j b_j\}$ . Assume  $a_1 \leq a_*$ , since otherwise no feasible book exists, and replace  $m$  by  $\min\{m, N\}$ . Restrict books to  $c_1 \leq a_*$ ; this retains exactly those books that can meet the root promise. At a real price  $\lambda$ , define

$$g_\lambda(x) = f(x) - \lambda x, \quad \phi_j(z; \lambda) = \max_{0 \leq y \leq z} \{-h_j(y) - \lambda y\}, \quad (43)$$

$$D(\lambda) = \lambda B + \max_{\emptyset \neq c \subset \mathcal{A}, |c| \leq m, c_1 \leq a_*} \left[ \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\} - \sum_{a_i \in c} \rho_i \right]. \quad (44)$$

Weak duality gives  $\mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq D(\lambda)$  in the renewal specialization. For quadratic shortfall costs, (43) is evaluated at  $y = 0$  if  $\lambda \geq 0$ , and at  $y = \min\{z, -\lambda/\gamma_j\}$  otherwise; when  $\gamma_j = 0$  and  $\lambda < 0$ , use  $y = z$ .

Call an arc  $u < v$  admissible when  $g_\lambda(v) \geq g_\lambda(u)$ . Define its reward and the reward for stopping at  $u$  by

$$E_\lambda(u, v) = \sum_{j: u \leq b_j < v} \pi_j \begin{cases} g_\lambda(u) + \frac{b_j - u}{v - u} [g_\lambda(v) - g_\lambda(u)], & v \leq \tau_j, \\ g_\lambda(u) + \phi_j(b_j - u; \lambda), & v > \tau_j, \end{cases} \quad (45)$$

$$T_\lambda(u) = \sum_{j: b_j \geq u} \pi_j [g_\lambda(u) + \phi_j(b_j - u; \lambda)]. \quad (46)$$

An edge accounts only for branches whose target caps it crosses. It therefore carries a contractual allocation value, not an ordinary nearest-neighbor quantization error.

**Theorem A.1** (Polynomial price decomposition). *For every  $\lambda$ , some maximizer in (44) has nondecreasing  $g_\lambda$  along its ordered book. Its priced operating value is the sum of (45) over consecutive levels plus (46) at*

the last level. Set

$$W_1(i) = T_\lambda(a_i), \tag{47}$$

$$W_r(i) = \max \left\{ T_\lambda(a_i), \max_{\substack{\ell > i \\ g_\lambda(a_\ell) \geq g_\lambda(a_i)}} [E_\lambda(a_i, a_\ell) - \rho_\ell + W_{r-1}(\ell)] \right\}.$$

Then  $D(\lambda) = \lambda B + \max_{a_i \leq a_s} [W_m(i) - \rho_i]$ . With constant-time evaluation of  $\phi_j$ , computing a maximizing book and this bound takes  $O((k+m)N^2)$  arithmetic operations and  $O(N^2 + mN)$  storage, allowing both  $k$  and  $m$  to vary. Rational quadratic data and a rational price give an exact rational algorithm with polynomial bit complexity. General costs use at most  $O(kN^2)$  calls to the scalar oracle (43) instead.

*Proof.* If  $\lambda < 0$ , strict increase of  $f$  makes  $g_\lambda$  strictly increasing. If  $\lambda \geq 0$ , intermediate service cannot improve priced payoff, and a branch maximizes the interpolant of  $g_\lambda$  on its eligible prefix, truncated at its target cap. Concavity implies that the values along any book increase up to their first maximum and never subsequently increase. All levels after that first maximum may be deleted: for a branch below the maximizing level, either that level is ineligible and all later levels are also ineligible, or later chords cannot improve the concave interpolant before the cap; for a branch above it, no later value beats the maximum. Nonnegative charges make deletion weakly beneficial. The retained book has only admissible arcs.

Consider  $b_j \in [u, v)$  on such a book. If  $v \leq \tau_j$ , the best priced target is  $b_j$  on the chord  $[u, v]$ . For  $\lambda \geq 0$  its slope is nonnegative and a singleton tail contributes no positive adjustment. For  $\lambda < 0$ , a singleton tail contributes at most  $-\lambda(b_j - u)$ , whereas the chord contributes this amount plus a strictly positive reward increment when  $b_j > u$ . If  $v > \tau_j$ , every later level is ineligible; the optimum is a singleton at  $u$  followed by the best shortfall in (43). Branches beyond the final level have the same singleton-tail form. These disjoint cap intervals cover all branches because the first level is at most  $\min_j b_j$ . This proves the decomposition, including equality cases.

The acyclic recurrence chooses either to stop or to append the next admissible level, paying each selected charge once. It enumerates paths implicitly rather than enumerating books. Computing edges costs  $O(kN^2)$ ; the dynamic-programming layers cost  $O(mN^2)$ . For rational quadratic data, every scalar oracle value uses a bounded number of rational operations, and each path value sums at most  $O(km)$  such terms. Numerator and denominator lengths are polynomial in the input bit length; comparisons and path reconstruction therefore preserve polynomial bit complexity.  $\square$

Pricing and constrained shortest-path bounds are established optimization mechanisms (Handler and Zang, 1980). The model-specific result here is that one price makes an otherwise nonseparable shared-book allocation decompose into these cap-crossing edges, despite heterogeneous shortfall costs, branch-specific eligibility, and nonsaturated targets. The theorem does not assert a polynomial algorithm for the unrelaxed joint problem or exchange its maximization with minimization over prices.

## A.2 Prefix bounds and checkable global search

A node is a mandatory initial segment  $p = (i_1, \dots, i_s)$  of a book; its descendants may append only larger indices. If the priced values along  $p$  never decrease, its exact priced completion value is

$$U_p(\lambda) = \lambda B + \sum_{h=1}^{s-1} E_\lambda(a_{i_h}, a_{i_{h+1}}) - \sum_{h=1}^s \rho_{i_h} + W_{m-s+1}(i_s). \quad (48)$$

If a decrease occurs in  $p$ , concavity places its first priced maximum within  $p$  and every larger new level is useless to the priced allocation. In that case, use the fixed-prefix branch-support value, with every mandatory charge retained. This is again the exact priced maximum over completions; additional nonnegative charges cannot help. The empty prefix uses (44).

**Theorem A.2** (Finite search and an anytime global interval). *Fix any nonempty finite set of prices. At each prefix use the smallest corresponding  $U_p(\lambda)$ , evaluate feasible books by the exact inner allocator, and retain the largest feasible net value  $L$ . Partition an unpruned node into its own book, when nonempty, and all one-index extensions. Prune a node only when its upper bound is at most  $L$ . This procedure terminates after finitely many prefixes and returns the exact catalog optimum. At any interruption it returns*

$$L \leq \mathcal{V}_{m,\mathcal{A}}^\delta(B; \rho) \leq U = \max\{L, \max_{p \text{ unresolved}} \min_{\lambda} U_p(\lambda)\}. \quad (49)$$

*For a uniform mesh of spacing  $h$ , Theorem E.1 further yields the continuous interval  $[L, U + K_m h]$ . The optimization gap  $U - L$  and discretization error  $K_m h$  are separate and observable.*

*Proof.* The preceding completion argument proves each price bound, so their minimum is valid by weak duality. Every feasible book either equals a nonempty node or has a unique next index. Hence the prescribed children and the node's own book are a disjoint exhaustive partition; book size is at most  $m$ . A pruned subtree cannot improve the incumbent. The maximum of unresolved upper bounds and the incumbent therefore bounds every feasible book at every stage. If all subtrees have been resolved, the two bounds coincide. The mesh transport is a separate one-sided inequality between the continuous optimum and the exact catalog optimum, so substituting the catalog upper bound proves the last claim.  $\square$

A global witness consists of the feasible winning lottery, every split and its complete child list, upper bounds and prices at pruned or unresolved nodes, and a fixed-book supporting price for each explicitly evaluated node book. A checker can rebuild price tables and verify the partition without replaying the search order or trusting the incumbent-selection code. This is a stronger object than an inner multiplier or a source hash. Equality of one root price bound with a feasible payoff already certifies global catalog optimality without any branching. Worst-case search is still exponential in the alphabet budget; neither this result nor the mesh transfer is an FPTAS claim.

**Additional decomposition contribution.** Theorems A.1–A.2 use classical Lagrangian bounds and an acyclic path recurrence. Their structural increment is the exact eligibility-aware cap-crossing reward and

prefix-completion dominance for nonsaturated allocation with heterogeneous shortfall costs. The generic search framework is not claimed as new. Proposition E.3 establishes sharpness for uniform catalogs, not a lower bound for adaptive grids. The continuous stationary-face mechanism remains generic; its contractual closed-cell reduction remains the structural contribution.

## B Exact Continuous Quadratic Design

Fix  $s \leq m$  and temporarily allow  $0 \leq c_1 \leq \dots \leq c_s \leq 1$ . For each branch select one of  $2s - 1$  modes. A singleton mode  $S_\ell$  and an adjacent-lottery mode  $L_\ell$  have the following closed descriptions:

$$S_\ell : \quad c_\ell \leq t_j \leq b_j, \quad g_j^S = rc_\ell - \frac{q}{2}c_\ell^2 - \frac{\gamma_j}{2}(t_j - c_\ell)^2; \quad (50)$$

$$L_\ell : \quad c_\ell \leq t_j \leq c_{\ell+1}, \quad t_j \leq b_j, \quad c_{\ell+1} \leq b_j + \delta, \quad g_j^L = rt_j - \frac{q}{2}[(c_\ell + c_{\ell+1})t_j - c_\ell c_{\ell+1}]. \quad (51)$$

For  $S_\ell$ , use  $y_j = t_j - c_\ell$  and terminal level  $c_\ell$ . For  $L_\ell$ , use  $y_j = 0$  and the adjacent unbiased lottery. At  $c_\ell = c_{\ell+1}$  the constraints force  $t_j = c_\ell$ , and the formula is the singleton reward. Therefore collisions are not discarded as singular probability formulas. The quadratic interpolation identity eliminates the ratio in the lottery probabilities before codewords become optimization variables.

Each joint mode assignment, the ordered-book inequalities, and  $\sum_j \pi_j t_j = B$  define a compact polytope. If  $\rho(c) = \alpha c^2 + \beta c + \eta$  is nonnegative on  $[0, 1]$ , its objective

$$Q(c, t) = \sum_j \pi_j g_j(c, t) - \sum_{\ell=1}^s \rho(c_\ell) \quad (52)$$

is a quadratic polynomial, not necessarily concave. Nonnegative charges, rather than signs of their coefficients, are the assumption. For example, a decreasing nonnegative certification charge is allowed.

**Lemma B.1** (Finite stationary-face search). *A quadratic function on a nonempty compact polytope attains its maximum at a point that is either a vertex or the unique stationary point on the affine hull of some face with negative-definite restricted Hessian. With rational input data, some maximizing point and its value are rational and can be found by finitely many rational linear solves and feasibility tests.*

*Proof.* Among all faces whose relative interiors contain a global maximizer, choose one of least dimension and a maximizing point  $x$  in its relative interior. The derivative along the face's tangent space vanishes and the restricted Hessian is negative semidefinite. If this restriction has a nonzero null direction  $z$ , the quadratic is constant along  $x + az$  in the affine hull: both its linear and quadratic terms in  $a$  vanish. Compactness makes this line reach a proper face while remaining feasible, contradicting the choice of dimension. The restriction is therefore negative definite unless the tangent space is zero dimensional.

Enumerate independent subsets of the defining inequalities, together with the equality constraints, and solve the stationarity equations on each associated affine space. On the face just identified the bordered stationarity matrix is nonsingular; at a vertex, the tangent space is empty and the same conclusion holds. Test feasibility in every original inequality and retain the largest objective among all feasible candidates.

Each retained candidate is feasible, and at least one is a global maximizer. Rational elimination gives the rationality claim. Singular bordered systems may be skipped because the argument guarantees a nonsingular maximizing face of smaller dimension; it does not assume nondegeneracy of the original polytope.  $\square$

**Theorem B.2** (Exact joint continuous design). *For rational quadratic primitives, rational caps and probabilities, rational  $B \in [0, \bar{b}]$  and  $\delta \geq 0$ , and a rational quadratic charge nonnegative on  $[0, 1]$ , the maximum of (52) over the closed cells (50)–(51), for every  $1 \leq s \leq m$ , equals  $\mathcal{V}_m^\delta(B; \rho)$ . The optimum is attained by a rational book, rational targets, and rational lottery probabilities. Deleting the lottery modes gives the exact deterministic frontier at every promise, with the same charges.*

*An exact algorithm enumerates at most  $(2s - 1)^k$  cells at size  $s$ . After eliminating one target, each cell has dimension at most  $s + k - 1$  and at most  $s + 1 + 4k$  inequalities. Enumerating its independent active sets and applying Lemma B.1 gives a finite global algorithm with at most*

$$\sum_{s=1}^m (2s - 1)^k 2^{s+1+4k} \quad (53)$$

*linear systems, each of polynomial dimension and rational encoding length in the input size. The sharper rank-limited count replaces  $2^{s+1+4k}$  by  $\sum_{r=0}^{\min\{s+k-1, s+1+4k\}} \binom{s+1+4k}{r}$ . A lottery branch may impose four separate inequalities: its two chord bounds, its target cap, and its realization eligibility bound. Neither of the latter two is redundant at a general positive tolerance. This is an exponential reference algorithm, not a polynomial-time claim or an extension of the saturated Monge complexity.*

*Proof.* Theorem 2.1 turns every optimal book and target allocation into singleton or adjacent-lottery branch modes represented in these cells. Conversely every cell point implements a feasible policy by the displayed reconstruction; all constraints, including the exact root promise, hold. If ordered slots collide or some slots are unused, deleting redundant slots cannot increase a nonnegative charge. The resulting smaller book and its canonical branch rules appear among the enumerated sizes. Hence overcounting charges at duplicate slots cannot change the maximum across all sizes. These two directions prove equality with the original optimization problem, not merely a relaxation or a fixed-book upper bound.

There are finitely many compact cells and each has a continuous quadratic objective. Lemma B.1 supplies a rational maximizing candidate on each nonempty cell. Adjacent distinct rational levels reconstruct rational probabilities; coincident levels reconstruct a singleton. The mode count, inequality count, and all-subsets upper bound give (53). Standard determinant bounds for rational linear systems give polynomial encoding length for each candidate. Singleton-only modes exactly represent deterministic terminal assignments and intermediate tiers, proving the deterministic statement.  $\square$

The reduction solves the unrestricted continuous outer problem at an interior promise, rather than relabeling fixed-book allocation as global design. Its contract-specific content is the eligibility-prefix reduction and the closed quadratic elimination of lottery probabilities with no loss at collisions. Rational quadratic optimization and active-set reasoning are classical; polynomial-size witnesses are not a new general complexity principle (Del Pia et al., 2017). The fast saturated Monge theorem remains valuable precisely because the general finite search is exponential.

### B.1 Pathwise comparison, repeated caps, and small charges

Theorem B.2 computes  $V_m^E(B)$ ,  $V_m^P(B)$ , and  $V_m^D(B)$  for every rational promise, using respectively all lottery modes at  $\delta = 1$ , all modes at  $\delta = 0$ , and singleton modes. No claim of pathwise-deterministic equality away from saturation is needed for this characterization. The complete predecessor companion retains the earlier exact institutional comparison path; the current study reports the separately executed budgeted-design comparisons. The zero-memory-loss interval  $B \leq \min b_j$  in Theorem EC.4.1 remains important: a positive institution gap is not expected throughout the feasible promise range.

**Proposition B.3** (Pathwise randomization can matter in the interior). *For one branch with  $b = 9/10$ ,  $B = 9/20$ ,  $r = 2$ ,  $q = \gamma = 1$ , two available symbols, and the nonnegative charge  $\rho(c) = 3c(1 - c)$ ,*

$$\mathcal{V}_2^0(B; \rho) = \frac{171}{400} > \mathcal{V}_2^D(B; \rho) = \frac{9}{160}.$$

*The pathwise optimum uses book  $(0, 9/10)$ , equal draw probabilities, and no intermediate tier. The deterministic optimum uses the single level  $9/20$ .*

*Proof.* A deterministic policy uses only one level  $0 \leq c \leq B$ ; deleting unused levels weakly lowers its charge. Its net payoff is  $-81/800 - (11/20)c + 2c^2$ , whose maximum over  $[0, B]$  is  $9/160$  at  $B$ . Every nonsingleton canonical pathwise mode has  $0 \leq u \leq B \leq v \leq b$ , and net payoff  $2B - (u+v)B/2 + uv/2 - 3u(1-u) - 3v(1-v)$ . This is convex in  $(u, v)$ , so its maximum on that rectangle is at a corner. Checking its four corners, together with the singleton case, gives  $(u, v) = (0, 9/10)$  and value  $171/400$ . Both realizations meet the hard cap and their mean is exactly  $B$ .  $\square$

This example uses a specified nonnegative, nonmonotone level charge; it is not a counterexample asserted for the uncharged problem. It shows why the saturated pathwise collapse cannot simply be imported into the unified interior design model.

**Proposition B.4** (Homogeneous repeated-cap aggregation). *For the randomized institutions in (1), branches with identical caps and identical intermediate-cost functions can be aggregated by adding their probabilities, at every  $B$ ,  $\delta$ , and codeword charge. This holds for each fixed book and after global book optimization.*

*Proof.* Such branches have identical eligible prefixes and the same concave response (2). Replacing their targets by their probability-weighted mean preserves the root promise and does not lower total reward by Jensen's inequality. Replicating an aggregate target gives the converse inequality. Charges depend only on the common book. This argument does not aggregate heterogeneous costs or establish a corresponding deterministic restriction.  $\square$

The strict off-cap conclusion is also stable under genuinely selected-level charges. In Proposition C.3, the uncharged two-level book  $(1/4, 5/8)$  beats every book of at most two cap levels by  $1/1280$ . If  $0 \leq \rho(c) \leq \eta$  on  $[0, 1]$  and  $\eta < 1/2560$ , evaluating this same off-cap book incurs less than that gap, while a cap-only book cannot improve its uncharged payoff after a nonnegative charge. Thus no charged optimum is cap-only. This statement preserves the off-cap property, not necessarily the numerical location  $5/8$ .

## C The Global Continuous Lottery Codebook

The fixed-codebook formula in Theorem EC.3.1 leaves a continuous design problem: both the terminal service levels and their assignment probabilities must be chosen. We now solve that problem globally for quadratic terminal rewards and heterogeneous quadratic intermediate costs. The solution also determines when imposing participation after, rather than before, the private draw eliminates the value of randomization. These are restricted-alphabet design results, distinct from the exact-implementation alphabet in Theorem EC.4.1.

Retain the three-date renewal architecture of Theorem EC.2.1, including the saturated root promise  $\bar{b} = \sum_{j=1}^k \pi_j b_j$ , strictly ordered caps  $0 < b_1 < \dots < b_k < 1$ , and positive probabilities summing to one. Throughout this section,

$$f(c) = rc - \frac{q}{2}c^2, \quad h_j(y) = \frac{\gamma_j}{2}y^2, \quad r \geq q > 0, \quad \gamma_j \geq 0. \quad (54)$$

The parameters and probabilities can differ across branches as indicated. Terminal reward is strictly increasing on the feasible cap range, including when  $r = q$ . All private information retained until the common terminal vertex must be encoded in the charged symbol. The decoder and public observation convention are unchanged.

### C.1 Cap anchors and one continuous terminal level

For  $u \leq b \leq v$ , quadratic concavity gives the interpolation loss

$$f(b) - \frac{v-b}{v-u}f(u) - \frac{b-u}{v-u}f(v) = \frac{q}{2}(b-u)(v-b), \quad (55)$$

with the endpoint value understood when levels coincide. Thus an interior codeword enters the loss affinely until it crosses a branch cap. The highest codeword is different: reducing it below a branch cap creates an intermediate payment and its associated cost.

**Lemma C.1** (Cap-anchored representation). *For every alphabet budget  $m \geq 1$ , an optimum of (EC.5) exists with the following representation. Its lowest codeword is  $b_1$ ; every codeword except the highest belongs to  $\{b_1, \dots, b_k\}$ ; and the highest belongs to  $[b_1, b_k]$ . A one-symbol optimum is  $\{b_1\}$ . The highest codeword need not be a branch cap.*

The proof is in Electronic Companion Section EC.5.4.

The lemma does not discretize the highest level. A cap-only restriction can be strictly suboptimal, as Proposition C.3 shows. It also does not assert that every optimal codebook is anchored; it supplies an optimum with this useful form.

## C.2 A finite exact algorithm

For adjacent anchored levels  $b_i < b_j$ , define

$$E(i, j) = \frac{q}{2} \sum_{h=i}^j \pi_h (b_h - b_i)(b_j - b_h). \quad (56)$$

Let  $P(s, j)$  be the minimum interpolation loss on caps through  $b_j$  using exactly  $s$  anchored levels, starting at  $b_1$  and ending at  $b_j$ . Set  $P(1, 1) = 0$ ,  $P(1, j) = +\infty$  for  $j > 1$ , and use the ordered recurrence

$$P(s, j) = \min_{i < j} \{P(s-1, i) + E(i, j)\}, \quad 2 \leq s \leq j. \quad (57)$$

This is a standard shortest-path or segmentation recurrence. The substantive reduction is Lemma C.1, which makes it applicable to a jointly optimized continuous lottery codebook.

Suppose that  $b_i$  is the last anchored level and  $z$  is the final level. For  $\ell \geq i$  and  $b_\ell \leq z \leq b_{\ell+1}$ , put  $b_{k+1} = b_k$  and define

$$\begin{aligned} \Psi_{i\ell}(z) = & \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)(z - b_h) \\ & + \sum_{h=\ell+1}^k \pi_h \left\{ f(b_h) - f(z) + \frac{\gamma_h}{2} (b_h - z)^2 \right\}. \end{aligned} \quad (58)$$

The first sum is the lottery loss below  $z$ ; the second is the terminal-reward and intermediate-payment loss above it. Empty sums are zero. For  $\ell < k$  the function is strictly convex in  $z$  and has the constrained minimizer

$$z_{i\ell} = \left[ \frac{r \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h b_h - \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)}{q \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h} \right]_{[b_\ell, b_{\ell+1}]}. \quad (59)$$

For  $\ell = k$ , take  $z_{ik} = b_k$ . Define the terminal cost

$$T_i = \min_{i \leq \ell \leq k} \Psi_{i\ell}(z_{i\ell}). \quad (60)$$

**Theorem C.2** (Global frontier and exact complexity). *Under (54), the optimum with at most  $m$  symbols is*

$$D_m^{\text{ran}} = \min \left\{ D_1, \min_{\substack{1 \leq s \leq \min(m-1, k) \\ 1 \leq j \leq k}} \{P(s, j) + T_j\} \right\}, \quad (61)$$

where the inner minimum is empty for  $m = 1$ . Backtracking produces a globally optimal codebook and the adjacent lotteries implementing it. For ordered inputs and  $1 \leq m \leq k$ , all frontiers through budget  $m$  can be computed in  $O(mk^2)$  exact rational operations and  $O(mk + k)$  stored rational numbers and indices, including backpointers. No codeword grid is required.

For rational input of total binary length  $L$ , a reduced-rational implementation has polynomial Turing

complexity. A conservative bound is  $O(mk^2S^3)$  bit operations with  $S = O(k[L + \log(k+2)])$ . This includes exact cost comparisons, clamping, and fraction reduction. For  $m \geq k$  the optimal loss is zero; listing additional budget entries requires only their output cost. These guarantees concern the stated algorithm, not an arbitrary nonlinear optimizer for (EC.5).

The proof is in Electronic Companion Section EC.5.5.

### C.3 A continuous optimal highest level

**Proposition C.3** (An off-cap terminal level is optimal). *Let  $(b_1, b_2, b_3) = (1/4, 1/2, 3/4)$ ,  $(\pi_1, \pi_2, \pi_3) = (7/20, 3/5, 1/20)$ ,  $r = 2$ ,  $q = 1$ , and  $\gamma_j = 1$ . The globally optimal two-symbol expected-participation codebook is  $(1/4, 5/8)$ , and*

$$D_2^{\text{ran}} = \frac{23}{1280} < \frac{3}{160} = \min_{\substack{|c| \leq 2 \\ c \subseteq \{b_1, b_2, b_3\}}} D(c) = D_2, \quad (62)$$

where the cap-only minimum is over feasible codebooks and  $D(c)$  denotes their expected-participation loss. Thus even optimizing all cap-only lotteries misses the unrestricted optimum.

The proof is in Electronic Companion Section EC.5.6.

### C.4 Participation after the private draw

The preceding frontier is appropriate when a renewal review approves an expected continuation before the protocol draws its private symbol. Consider instead an agreement that requires the same cap to hold for every private realization after that review. A draw-specific intermediate tier is allowed; only the terminal public state and the charged symbol reach the terminal decoder. Write  $D_m^{\text{pw,ran}}$  for the best randomized loss under these pathwise constraints and the same saturated root promise.

**Theorem C.4** (Pathwise randomization has no frontier advantage). *For the general renewal family of Theorem EC.2.1, with strictly increasing concave  $f$  and convex nondecreasing  $h_j$ , and the observation architecture just specified,*

$$D_m^{\text{pw,ran}} = D_m \quad \text{for every } m \geq 1. \quad (63)$$

Consequently (61) and the deterministic frontier give an exact comparison of the two participation institutions under (54). Their exact-optimum alphabet thresholds are both  $k$  when strict concavity makes the  $k$  distinct full-information actions unique.

The proof is in Electronic Companion Section EC.5.7.

This institutional result is not an assertion that randomization is generally ineffective in dynamic contracts. It uses the saturated root promise, an exogenous recombination, and the charged-information interface. Under expected participation, Theorem C.2 chooses a less expensive lottery protocol; under pathwise participation, the same service model calls for the deterministic frontier. In either institution an operator can price a symbol budget, add its implementation charge to the computed loss, and optimize the budget without confusing read-only program size with retained private information.

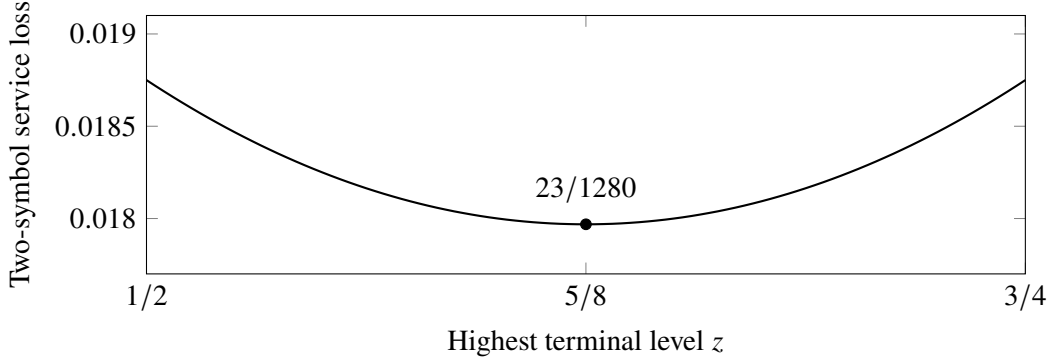


Figure 1: The analytically optimized highest level in the heterogeneous three-branch example. The lower level is  $1/4$ . The displayed function is  $z^2/20 - z/16 + 3/80$  on  $[1/2, 3/4]$ ; its interior minimum is  $z = 5/8$ , not a branch cap.

## D Monge Costs and Classical Linear Matrix Search

The finite dynamic program still optimizes a continuous highest service level. We first prove the contractual cost inequalities and then verify the classical easy-direction completion with the exact order used by the code. A search routine cannot establish the economic reduction that supplies its entries.

### D.1 The inequality survives continuous terminal minimization

Write  $I_\ell = [b_\ell, b_{\ell+1}]$ , including  $I_k = \{b_k\}$ , and  $Q(i, \ell) = \min_{z \in I_\ell} \Psi_{i\ell}(z)$  for  $i \leq \ell$ . A partial array is Monge here when

$$A(r, c) + A(s, d) \leq A(r, d) + A(s, c), \quad r < s, \quad c < d, \quad (64)$$

whenever all four entries are defined. This fixes the inequality convention before any orientation is named.

**Lemma D.1** (Contractual finite-cost inequalities). *The anchored edge costs  $E$  and deterministic cell costs  $C$  satisfy  $A(a, c) + A(b, d) \leq A(a, d) + A(b, c)$  for  $a \leq b \leq c \leq d$ . The continuously minimized terminal array  $Q$  is Monge on its feasible domain. These statements permit arbitrary positive probabilities and heterogeneous nonnegative intermediate curvatures.*

The proof is in Electronic Companion Section EC.5.8.

**Lemma D.2** (Previous-layer values are column offsets). *Adding any finite column term  $v_c$  preserves the Monge inequality. For fixed finite predecessor values, the arrays  $A(j, i) = P(s, i) + E(i, j)$  and  $B(j, i) = D(s - 1, i) + C(i + 1, j)$  are therefore Monge wherever all four entries are feasible.*

The proof is in Electronic Companion Section EC.5.9.

### D.2 Completion with the implemented strict and tie orders

Give a feasible entry the lexicographic key

$$K(r, c) = (0, A(r, c), c). \quad (65)$$

Finite ties select the smaller column, and distinct columns have distinct keys. The exact total-monotonicity condition used by the implementation is

$$K(r, d) < K(r, c) \implies K(s, d) < K(s, c), \quad r < s, c < d. \quad (66)$$

A strict preference for the right column cannot reverse in a later row. This applies to arbitrary selected rows and columns. Because keys are distinct, a decrease of row-argmin indices in any submatrix would contradict (66).

**Lemma D.3** (Suffix-feasible completion). *Let row  $r$  have feasible columns  $c \geq \ell_r$ , with nondecreasing cutoffs  $\ell_r$ . Suppose the finite entries are Monge. Complete every  $c < \ell_r$  with  $K(r, c) = (1, 0, -c)$ . Then (66) holds. A completely infeasible row of a selected submatrix selects its largest selected column.*

The proof is in Electronic Companion Section EC.5.10.

**Lemma D.4** (Prefix-feasible completion). *Let row  $r$  have feasible columns  $c \leq u_r$ , with nondecreasing cutoffs  $u_r$ . Suppose the finite entries are Monge. Complete every  $c > u_r$  with  $K(r, c) = (1, 0, c)$ . Then (66) holds. A completely infeasible selected row selects its smallest selected column.*

The proof is in Electronic Companion Section EC.5.11.

These lemmas specialize the known favorable completion setting in Section 1; they are not general partial-matrix search theorems. A helper accepting two bounds does not prove total monotonicity for arbitrary interval rows. For example, two zero-cost rows feasible only at columns 2 and 1 have a vacuously Monge finite part but decreasing minima. Uniform infinite padding with ordinary left ties also fails on the required suffix domain: rows with cutoffs 2, 3 restricted to columns 1, 2 select 2, 1. The opposite internal padding orders above are deliberate.

### D.3 Reachability and three search interfaces

**Lemma D.5** (Contiguous reachable predecessor sets). *For the anchored program,  $P(1, j)$  is finite exactly at  $j = 1$ , and for  $s \geq 2$ ,  $P(s, j)$  is finite exactly when  $s \leq j \leq k$ . For the deterministic program,  $D(0, j)$  is finite only at  $j = 0$ , and  $D(s, j)$  is finite exactly when  $s \leq j \leq k$  for  $s \geq 1$ . Each globally reachable predecessor set that can extend to a further endpoint is a contiguous interval or the initial singleton.*

The proof is in Electronic Companion Section EC.5.12.

The following pseudocode uses one-based cap indices and zero for the empty deterministic prefix. “Search” means ordinary full-matrix SMAWK on the supplied key. All prior-layer values are known before a search.

**Terminal interface.** Rows and columns are  $1, \dots, k$ . At  $(i, \ell)$  return  $(1, 0, -\ell)$  if  $\ell < i$ ; otherwise minimize (58) by (59) and return  $(0, Q(i, \ell), \ell)$ . Search all rows. Reevaluate each winning cell once to store  $T_i$  and its continuous optimizer.

**Anchored-prefix interface.** To form layer  $s+1$ , use rows  $j = s+1, \dots, k$ . Columns are  $\{1\}$  for  $s = 1$ , and  $s, \dots, k-1$  otherwise. At  $(j, i)$  return  $(1, 0, i)$  if  $i \geq j$ , and  $(0, P(s, i) + E(i, j), i)$  otherwise. Search and

store values and predecessors. Scan the current reachable endpoints with  $T_i$  to update the at-most-budget incumbent; backtrack only its winning codebook.

**Deterministic-prefix interface.** To form layer  $s$ , use rows  $j = s, \dots, k$ . Columns are  $\{0\}$  for  $s = 1$ , and  $s - 1, \dots, k - 1$  otherwise. Return  $(1, 0, i)$  if  $i \geq j$ , and  $(0, D(s - 1, i) + C(i + 1, j), i)$  otherwise. Search, store predecessors, and backtrack the solution ending at  $k$ .

Lemma D.5 justifies every discarded column. The terminal interface has shrinking feasible suffixes; the program interfaces have expanding feasible prefixes. The completion lemmas include infeasible rows in submatrices created internally by SMAWK, although every original searched row has a feasible entry. Padded comparisons do not evaluate an undefined economic cost.

**Theorem D.6** (Contractual corollary of classical matrix search). *For ordered rational inputs satisfying (54) and  $1 \leq m \leq k$ , both expected-participation and deterministic/pathwise frontiers through budget  $m$ , including an optimal codebook for every budget, require  $O(mk)$  rational operations and comparisons and  $O(mk + k)$  stored rational numbers and indices. The highest randomized level remains continuous. With total input length  $L$  and  $S = O(k[L + \log(k + 2)])$ , a conservative Turing bound is  $O(mkS^3)$ . Sorting unordered distinct caps adds  $O(k \log k)$  exact comparisons.*

The proof is in Electronic Companion Section EC.5.13.

The retained divide-and-conquer implementation is a valid  $O(mk \log(k + 1))$  alternative using ordinary row-minimum monotonicity. It is an implementation comparator rather than another headline theorem.  $O(k)$  exact-rational oracle/comparison work per layer does not imply uniform wall-time superiority for a particular Python implementation.

## E Catalog Approximation Without Promise or Risk Slack

Finite catalogs have two different interpretations. An externally certified menu is a restriction of the feasible technology, for which an exact catalog optimum is the relevant answer. Alternatively the operator may choose a numerical catalog to approximate continuous design. The latter interpretation needs a quantitative comparison to the unrestricted optimum; exactness on the catalog alone does not supply it.

Let  $\mathcal{A} \subset [0, 1]$  contain zero and one, and let  $h$  be its largest consecutive spacing. Assume  $f$  is increasing, concave, and  $L_f$ -Lipschitz, and each  $h_j$  is convex, nondecreasing, and  $H_j$ -Lipschitz on  $[0, b_j]$ . Let the nonnegative charge  $\rho$  be  $L_\rho$ -Lipschitz on  $[0, 1]$ . Define

$$K_m = L_f + \sum_j \pi_j H_j + m L_\rho. \quad (67)$$

For the quadratic model, one can take  $L_f = r$  and  $H_j = \gamma_j b_j$ . Neither the root promise nor the risk tolerance is discretized.

**Theorem E.1** (Uniform exactly feasible catalog certificate). *For every  $B \in [0, \bar{b}]$ , every  $\delta \geq 0$ , and every  $m \geq 1$ ,*

$$0 \leq \mathcal{V}_m^\delta(B; \rho) - \mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq K_m h. \quad (68)$$

More strongly, every canonical continuous-book policy has a catalog policy with no more symbols that preserves each branch target exactly, satisfies the same realization ceiling exactly, and loses at most  $K_m h$  in net payoff. The conclusion includes the pathwise endpoint  $\delta = 0$  and requires no strict feasibility margin.

*Proof.* Round each original level  $c$  downward to  $a(c) = \max\{a \in \mathcal{A} : a \leq c\}$  and merge collisions. Fix a branch target  $t$  and let  $d$  be its old eligible top. Every old eligible level remains eligible after rounding. In particular the new eligible top  $d'$  satisfies  $d' \geq a(d) \geq d - h$ . The old canonical terminal mean is  $\mu = \min(t, d)$ ; coupling its draws to their rounded levels gives a feasible-support lottery of mean  $\tilde{\mu} \in [\mu - h, \mu]$  and expected terminal reward at least the old reward minus  $L_f h$ .

Do not retain those coupled probabilities. Recompute the canonical lottery on the rounded eligible prefix at the same target  $t$ . Its terminal mean  $\mu' = \min(t, d')$  is at least  $\tilde{\mu}$ : every rounded old draw is at most  $d'$  and its mean is at most  $t$ . The new interpolant is increasing, and at mean  $\tilde{\mu}$  it dominates the coupled lottery reward. Therefore its reward at  $\mu'$  also dominates that reward. Moreover  $\mu' \geq \mu - h$ , so its intermediate tier  $t - \mu'$  exceeds the old tier by at most  $h$ . The branch operating-payoff loss is at most  $(L_f + H_j)h$ . Theorem 2.1 ensures exact mean and risk feasibility for this reoptimized lottery, including when the risk bound is binding. All  $t_j$  are unchanged, so the root promise remains exactly  $B$ .

For each original level,  $\rho(a(c)) \leq \rho(c) + L_\rho h$ . After merging, the total charge is no larger than the sum of these rounded-slot charges because charges are nonnegative. At most  $m$  slots were present. Summing the branch bounds and charge bound gives  $K_m h$ . Apply the construction to an optimizer, or to a sequence approaching the supremum, and use inclusion of catalog policies in continuous policies for the other inequality.  $\square$

One-sided rounding is necessary but not sufficient without the probability repair. For example, let  $b = t = 1/2$ ,  $\delta = 1/8$ , and old support  $(1/4, 5/8)$  with upper probability  $2/3$ . Rounding the lower level to zero while retaining the probabilities reduces the mean to  $5/12$ . Compensating by  $y = 1/12$  produces a maximum obligation  $17/24 > 5/8$ , violating the risk limit. Instead, the new canonical lottery on  $(0, 5/8)$  uses upper probability  $4/5$  and  $y = 0$ . It preserves the target and the ceiling exactly. This is the distinction between unconstrained quantizer rounding and the contractual transport in Theorem E.1.

**Corollary E.2** (Certified approximation for a fixed alphabet budget). *For rational quadratic operating primitives and a nonnegative charge whose rational catalog evaluations and Lipschitz bound are supplied, enumerate every feasible catalog book of size at most  $m$  and solve its allocation by (4). With  $N = |\mathcal{A}|$ , this gives an exact catalog optimum in*

$$O\left(\left[\sum_{s=1}^{\min(m, N)} \binom{N}{s}\right] (km + k \log(k+1))\right) \quad (69)$$

*rational arithmetic/evaluation operations. A uniform mesh with denominator  $\lceil K_m/\varepsilon \rceil$  (or one when that ceiling is zero) gives additive error at most  $\varepsilon$ . The algorithm is polynomial in  $k$ ,  $K_m/\varepsilon$ , and the bit complexity of the sampled data for fixed  $m$ ; it is not claimed to be polynomial in a variable  $m$ , nor is this a multiplicative approximation ratio.*

*Proof.* Increasing subsets exhaust the allowed books and the supporting-price algorithm optimizes every inner problem exactly. Take the largest charge-adjusted value and apply Theorem E.1. The mesh includes zero, so every feasible promise is implementable even when it is smaller than the first positive grid point. The bound counts evaluation work; a black-box charge oracle is not assumed to have unit bit complexity unless that is explicitly part of the input model.  $\square$

The guarantee is uniform in the two contractual constraints. For any institution pair, subtracting their certified value intervals gives a certified interval for the institutional gap. This permits a global accuracy statement away from saturation without presuming that original cap anchors or Monge order survive. When  $B = \bar{b}$ , the faster charged path recurrence in Section EC.9 replaces subset enumeration. When the saturated expected problem is uncharged, the continuous Monge algorithm avoids a mesh entirely. Thus the specialized algorithms are faster members of one feasible-policy hierarchy, while (68) connects numerical catalogs to continuous risk-and-charge design.

### E.1 The uniform-mesh rate is sharp

The linear bound in Theorem E.1 reflects exact feasibility, not only an insensitive Lipschitz estimate. It cannot be uniformly replaced by a higher-order rate for the declared uniform catalogs.

**Proposition E.3** (A fixed-input lower bound). *Take one branch and one symbol,  $b = B = 1/2$ ,  $f(c) = c - c^2/2$ ,  $h(y) = y^2/2$ , and zero charges. For every risk tolerance and every odd integer  $N \geq 3$ , the uniform catalog with spacing  $h = 1/N$  has loss*

$$\mathcal{V}_1^\delta(B; 0) - \mathcal{V}_{1, \mathcal{A}_N}^\delta(B; 0) = \frac{h}{4} + \frac{h^2}{4}.$$

*Thus the worst-case  $O(h)$  rate is order-sharp, even for a single fixed input with no opening charges and no randomization.*

*Proof.* With one level,  $c \leq B$  and  $y = B - c$ . Payoff  $f(c) - (B - c)^2/2$  is strictly increasing on  $[0, B]$ . Its continuous maximizer is  $B$ , with value  $3/8$ . For odd  $N$ , the largest feasible catalog point is  $B - h/2$ . Substitution gives the displayed loss. All realizations equal total service  $B$ , so risk tolerance plays no role. The sequence of odd  $N$  uses the same primitives throughout.  $\square$

This is a lower bound for uniform catalogs, not for all adaptive discretizations. Adding the cap  $1/2$  removes this particular error. It also does not rule out faster convergence on other fixed instances. The search interval in (49) is therefore useful even when its optimization gap has already vanished: refinement addresses a genuinely different source of error.

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Table 1: Theorem-level contribution map. Antecedent statements and proofs remain in the current readers.

| Component                                   | Scope                                                                               | Provenance                                        |
|---------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------|
| Canonical and continuous design             | Exact response, institutional endpoints, continuous and Monge results               | Antecedent, Theorems 2.1, D.6                     |
| Saturated charged path                      | Globally exact forced targets                                                       | R42; Corollary 3.2                                |
| Unrestricted price path and prefix search   | Polynomial oracle and finite joint search                                           | R43; Theorems A.1–A.2                             |
| Conditional compression and full net        | Fixed targets; fixed-history additive scheme                                        | R45; Theorems 3.1, EC.12.2                        |
| Exact types, two-sided paths and box covers | Exact repeated-type reduction; polynomial box oracle; finite accuracy               | R46; Proposition 4.1, Theorems 5.1, 6.1           |
| Guarded response coarsening                 | Different caps and costs; explicit original-contract loss and lift                  | R47, Theorem 7.1                                  |
| Variable-history additive scheme            | Bounded eligibility and primitive scales; explicit accuracy dependence              | R47, Theorem 7.3                                  |
| Certified refinement and checking           | Nested valid intervals; original policy, grouping, Bellman and cover checks         | R47 refinement, Proposition 7.4; extended checker |
| Harmonic pooling and square-root bins       | Dominating upper model; optimal original lift; smaller accuracy-dependent dimension | R48, Theorems 8.1–8.4                             |

Table 2: Joint design at absolute tolerance 0.001. Counts include every frozen holdout case. Both methods optimize targets and books; neither column is conditional optimization.

| Branches | Catalog | Cases | Box completed | Prefix completed | Box seconds | Box max gap |
|----------|---------|-------|---------------|------------------|-------------|-------------|
| 2        | 9       | 4     | 4             | 4                | 0.018       | 0.00057     |
| 2        | 17      | 4     | 4             | 4                | 0.057       | 0.00010     |
| 4        | 9       | 4     | 4             | 4                | 0.028       | 0.00023     |
| 4        | 17      | 4     | 4             | 4                | 0.093       | 0.00094     |
| 8        | 9       | 4     | 3             | 4                | 0.106       | 0.01248     |
| 8        | 17      | 4     | 2             | 4                | 4.103       | 0.00673     |
| 128      | 33      | 4     | 3             | 4                | 0.398       | 0.00229     |
| 256      | 33      | 2     | 2             | 2                | 0.306       | 0.00078     |
| 16       | 33      | 2     | 2             | 2                | 0.850       | 0.00075     |

Seconds are medians for box optimization, excluding independent verification. The price-prefix baseline uses the inherited R43 completion bound, not merely an unpriced catalog relaxation. The 128- and 256-branch rows use exactly repeated types; the 128-branch row contains both four- and eight-type cases. Node and time limits, and all individual outcomes, are reported in the companion and machine-readable records.

Table 3: Completed and interrupted target grids versus adaptive joint design on identical finite catalogs. All have  $N = 5$  and  $m = 2$ . A dash means that the grid does not certify the requested accuracy.

| Branches | Tolerance | Grid profiles | Grid seconds | Grid certified | Box nodes | Box gap |
|----------|-----------|---------------|--------------|----------------|-----------|---------|
| 2        | 0.100     | 7             | 0.005        | Yes            | 1         | 0.00000 |
| 2        | 0.010     | 54            | 0.033        | Yes            | 1         | 0.00000 |
| 2        | 0.001     | 527           | 0.311        | Yes            | 1         | 0.00000 |
| 3        | 0.100     | 1000          | 0.826        | –              | 1         | 0.00003 |
| 3        | 0.010     | 1000          | 0.904        | –              | 1         | 0.00003 |
| 3        | 0.001     | 1000          | 0.908        | –              | 1         | 0.00003 |
| 4        | 0.100     | 1000          | 1.006        | –              | 1         | 0.01354 |
| 4        | 0.010     | 1000          | 1.063        | –              | 3         | 0.00903 |
| 4        | 0.001     | 1000          | 1.086        | –              | 1001      | 0.00128 |
| 5        | 0.100     | 1000          | 1.192        | –              | 1         | 0.00001 |
| 5        | 0.010     | 1000          | 1.163        | –              | 1         | 0.00001 |
| 5        | 0.001     | 1000          | 1.172        | –              | 1         | 0.00001 |

The grid has a declared cap of 1,000 evaluated profiles; these are profile-limited comparisons, not claims of time-matched speedup. Adaptive runs use at most 1,001 nodes and an eight-second soft limit. Exact values, objective scales, losses, and runtimes for both algorithms appear in the archived records.

Table 4: Distinct-history joint design at original-space tolerance 0.001. Every declared case is included. Histories in these rows have distinct cap–cost pairs.

| Histories | Catalog | Cases | Coarsened completed | Raw boxes completed | Priced prefix completed | Groups |
|-----------|---------|-------|---------------------|---------------------|-------------------------|--------|
| 4         | 9       | 4     | 4                   | 4                   | 4                       | 4      |
| 8         | 9       | 4     | 4                   | 2                   | 4                       | 5      |
| 12        | 9       | 4     | 4                   | 4                   | 4                       | 5      |
| 32        | 17      | 4     | 4                   | 3                   | 4                       | 5      |
| 128       | 17      | 4     | 4                   | 3                   | 3                       | 5      |
| 256       | 17      | 4     | 4                   | 3                   | 2                       | 5      |

The twelve primary cases use four, eight, or twelve histories, nine catalog levels, and two or three commands. The twelve scaling cases use seventeen catalog levels and three commands. The reduced solve uses tolerance 0.0005; completion above refers only to the checked original interval. Both box methods use at most 101 nodes and a four-second soft limit checked after full nodes. The priced-prefix comparator has the same requested original tolerance, node limit, and soft limit. All raw intervals and all lifted intervals pass their independent checkers.

Table 5: Optimization time and original-space gaps on all distinct-history cases. Seconds are local medians; maximum gaps are exact certificate widths rounded for display.

| Histories | Coarsened<br>seconds | Raw box<br>seconds | Prefix<br>seconds | Coarsened<br>max gap | Raw box<br>max gap | Prefix<br>max gap |
|-----------|----------------------|--------------------|-------------------|----------------------|--------------------|-------------------|
| 4         | 0.027                | 0.023              | 0.037             | 0.00034              | 0.00034            | 0.00081           |
| 8         | 0.246                | 0.684              | 0.069             | 0.00052              | 0.00212            | 0.00092           |
| 12        | 0.010                | 0.019              | 0.017             | 0.00010              | 0.00000            | 0.00000           |
| 32        | 0.618                | 0.372              | 0.315             | 0.00078              | 0.00261            | 0.00070           |
| 128       | 0.049                | 1.308              | 1.123             | 0.00013              | 0.00965            | 0.00965           |
| 256       | 0.095                | 3.981              | 3.503             | 0.00006              | 0.00173            | 0.00173           |

Timing excludes independent checking; coarsened timing includes reduction and original-space lifting. It is not uniformly faster: per-case outcomes, actual times above the soft limit, node counts, objective scales, and normalized gaps are retained in `results/joint.json`. The current code uses exact rational arithmetic. The data are a declared four-cluster perturbation family, not arbitrary-population or field evidence.

Table 6: Original-budget comparison across six seeds

| Charges    | Cases | Safeguarded exact |           | Core wins | Greedy wins | Max core loss |
|------------|-------|-------------------|-----------|-----------|-------------|---------------|
|            |       | Core exact        | in sample |           |             |               |
| None       | 18    | 18                | 18        | 2         | 0           | 0.000000      |
| Nonuniform | 18    | 17                | 18        | 8         | 1           | 0.028056      |

Exact means equality with the exhaustive joint catalog reference. Core and greedy wins compare their same-budget values; ties are omitted. The safeguarded policy keeps the better feasible candidate. The charged exception remains in the data. These counts are observations, not global guarantees for a finite target portfolio.

Table 7: Completed same-budget target-net comparisons

| Branches | Tolerance | Cases | Profiles | Maximum actual loss |
|----------|-----------|-------|----------|---------------------|
| 2        | 1/4       | 4     | 28       | 0.000000            |
| 2        | 1/8       | 4     | 50       | 0.000000            |
| 2        | 1/16      | 4     | 96       | 0.000000            |
| 3        | 1/2       | 4     | 47       | 0.049805            |
| 3        | 1/4       | 4     | 164      | 0.034645            |

All completed nets meet the requested additive tolerance with the original two-symbol budget and nonuniform charges. Counts aggregate four seeds. An additional interrupted run withholds its global accuracy guarantee.

Table 8: Conditional frontier scaling (targets fixed) with heterogeneous realization ceilings

| Seed   | Branches | Catalog | Budget | Charges | Selected | Seconds | Gap      |
|--------|----------|---------|--------|---------|----------|---------|----------|
| 104729 | 32       | 17      | 4      | No      | 4        | 0.036   | 0.019173 |
| 104729 | 32       | 17      | 4      | Yes     | 4        | 0.036   | 0.037009 |
| 104729 | 128      | 33      | 8      | No      | 8        | 0.486   | 0.004410 |
| 104729 | 128      | 33      | 8      | Yes     | 5        | 0.484   | 0.041534 |
| 104729 | 256      | 65      | 8      | No      | 8        | 3.257   | 0.004923 |
| 104729 | 256      | 65      | 8      | Yes     | 8        | 3.278   | 0.025671 |
| 205019 | 32       | 17      | 4      | No      | 4        | 0.038   | 0.010627 |
| 205019 | 32       | 17      | 4      | Yes     | 3        | 0.038   | 0.040935 |
| 205019 | 128      | 33      | 8      | No      | 8        | 0.474   | 0.003661 |
| 205019 | 128      | 33      | 8      | Yes     | 5        | 0.478   | 0.042000 |
| 205019 | 256      | 65      | 8      | No      | 8        | 3.303   | 0.004099 |
| 205019 | 256      | 65      | 8      | Yes     | 7        | 3.316   | 0.023976 |

Seconds are medians of three repeated executions on the same input. Gap is the valid same-budget Jensen upper bound minus the ideal-target conditional policy value, not an observed error to an unknown global optimum. Full timing arrays, stored-entry bit lengths and cumulative process resident-memory records are archived.

Table 9: Adaptive candidates at an unchanged installed-symbol budget

| Case           | Candidate pool | Candidates | Symbols | Exact-reference loss |
|----------------|----------------|------------|---------|----------------------|
| interior-ideal | uniform        | 9          | 2       | 0.044030             |
| interior-ideal | uniform        | 65         | 2       | 0.002991             |
| interior-ideal | oracle pool    | 4          | 2       | 0.000000             |
| off-cap-top    | uniform        | 9          | 2       | 0.000000             |
| off-cap-top    | Adaptive       | 5          | 2       | 0.000781             |
| off-cap-top    | Adaptive       | 6          | 2       | 0.000000             |
| Sat. 104729    | uniform        | 9          | 3       | 0.006507             |
| Sat. 104729    | Adaptive       | 9          | 3       | 0.000000             |
| Sat. 205019    | uniform        | 9          | 3       | 0.001824             |
| Sat. 205019    | Adaptive       | 9          | 3       | 0.000000             |

Every row has an exact continuous reference in the stated Jensen-attaining or saturated expected-participation regime. The interior oracle pool explicitly inserts ideal targets and is a sanity check. Candidate-pool size is not installed memory. Adaptive midpoint selection is a heuristic; catalog optimization at the saturated target is exact. The complete 22-row path is in the companion and JSON.

Table 10: Harmonic versus minimum-curvature aggregation at identical partitions and search limits. Entries report original-space certificates at tolerance 0.001, not reduced-solver completion. Every family contains six instances, including 32, 128, and 256 histories.

| Family                     | Certified / 6 |         | Maximum gap |           | Median seconds |          |
|----------------------------|---------------|---------|-------------|-----------|----------------|----------|
|                            | Harmonic      | Minimum | Harmonic    | Minimum   | Harmonic       | Minimum  |
| Common caps                | 6             | 3       | 0.0005123   | 0.009995  | 0.01367        | 0.01547  |
| Dispersed caps             | 1             | 1       | 0.0316      | 0.03165   | 0.1842         | 0.1223   |
| Eligibility / high promise | 1             | 1       | 0.003552    | 0.009106  | 0.01772        | 0.007898 |
| Zero cost / charged        | 6             | 6       | 0.0005434   | 0.0004811 | 0.04613        | 0.02399  |

Both methods receive 31 target boxes and four price refinements. Larger exact rational denominators and optimal original-group recovery can make harmonic pooling slower. A tighter reduced optimum is not a theorem about interrupted-search runtime or every returned interval.